

Black Holes as Constraint-Forced Origins

The Counter-Boundary Universe Theorem under AASC

AMetric Homology, No Inheritance, Constraint Symmetry, and Fresh Internal Governed Domains
Beyond Black-Hole Endpoints

AASC Black-Hole Endpoint Arc, Paper IV: Counter-Boundary Capstone

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Abstract

This manuscript proves the precise AASC sense in which certified AASC black-hole endpoints force fresh internal governed-domain status. A certified AASC black-hole endpoint is the post-subtraction target satisfying the unresolved-residue, maximum-governance-depth, New AMetric Boundary, no-crossing, and route-closure antecedents: the declared exterior quotient and every genuine certificate have removed exactly what they re-fix, leaving a nonempty unresolved residue. The theorem is universal over that certified target class; an astronomical label, a stationary representative, or a hair slogan neither proves nor evades the endpoint handoff. Such endpoints do not do so by inheritance, transport, tunneling, causal production, energy transfer, information transfer, copied law, or continuation of parent metricity. They expose a New AMetric Boundary with the same complete AMetric normal-form profile as the origin-side licensing boundary.

The central result is the Counter-Boundary Universe Theorem. The AMetric Normal-Form Homology theorem establishes

$$B_A^\star \neq B_A^0, \quad \text{PANF}(B_A^\star) = \text{PANF}(B_A^0) = \text{aANF}.$$

Homology alone gives no crossing, inheritance, cyclicity, causal production, or populated successor. The present theorem begins at that handoff and asks what counter-boundary status remains once the homologous endpoint boundary is fixed, no-crossing is preserved, admitted nullity is closed, and every non-fresh boundary-authority route is exhausted. The answer is exactly one admissible counter-boundary status-class up to $\equiv_{\text{gov}, CB}$: a fresh internal governed domain satisfying the AASC kernel under the AMetric normal form.

The resulting domain is role-equivalent to any origin-licensed governed-interior architecture at the level of minimal role structure, not because any law, metric, field, particle, energy ledger, information package, record, source ledger, clock, causal chain, parameter, or history crosses the boundary, but because the constraints are the minimal conditions on non-degenerate governed domainhood. No inheritance occurs because inheritance is inadmissible at an AMetric boundary. No-inheritance is the theorem's positive condition: it fixes the result as fresh constraint recurrence rather than transmissive succession. The unconditional theorem-level result is fresh internal governed-domain status; realized-physical and standard-physics contents remain target-class conditional and non-inherited.

Moreover, non-inheritance does not make the fresh domain arbitrary. If the fresh domain is instantiated as a realized physical interior, it must satisfy the same minimal role architecture; if the minimal Lorentzian metric-projection or realized-standard-physics target classes are fixed, the corresponding structural envelopes recur up to admissible redescription. Further downstream target classes are handled only under their own audited antecedents; the manuscript does not infer biological population, intelligence, technology, empirical abundance, or any concrete occupant from

the black-hole theorem. Numerical constants, histories, records, law tokens, parameter values, and concrete contents remain non-inherited, but any such contents must be internally certified by the new domain's own non-degeneracy conditions.

The proof is eliminative and constraint-theoretic rather than dynamical or transmissive. It is not a baby-universe tunneling model, cyclic cosmology, information-transfer theory, energy-transfer theory, or parent-to-child production mechanism. The controlled informal summary is: black holes make universes. The exact AASC statement is: certified AASC black-hole endpoints force fresh internal governed domains by constraint recurrence, not by inheritance. What is unique is the governance class up to $\equiv_{\text{gov},CB}$, not a selected token cosmology, populated history, or parent-derived empirical inventory.

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AASC Black-Hole Endpoint Arc Position

Shared arc spine: kernel/AMetric invariant $\rightarrow$ licensed AASC-GR black-hole regime $\rightarrow$ exterior no-totalization $\rightarrow$ standingless residual interior $\rightarrow$ governance-depth exhaustion $\rightarrow B_A^ \rightarrow$ AMetric homology $\rightarrow$ fresh internal governed domain.*

Shared series abstract. The AASC Black-Hole Endpoint Arc is a five-manuscript sequence that treats black holes as a fixed-domain stress test for AASC. It first role-types the licensed black-hole regime, then proves the exterior no-totalization obstruction, then classifies the uncertified horizon-isolated carrier as residual coherence rather than ordinary standing-bearing interiority, then exposes the endpoint AMetric normal form and no-crossing homology, and finally classifies the counter-boundary survivor as fresh internal governed-domain status. The sequence is conditional and one-way: later endpoint or capstone conclusions never authorize earlier target formation.

Paper label. Paper IV / Counter-boundary capstone. This manuscript is the only arc paper that states the controlled fresh internal governed-domain result.

This manuscript is Paper IV and the capstone of the arc. It is the only paper in the sequence that states the controlled fresh internal governed-domain result, and it states that result only in the exact AASC sense: certified endpoints force fresh internal governed-domain status by constraint recurrence.

Imported into this paper. The paper imports AMetric normal-form homology as a completed handoff: $B_A^* \neq B_A^0$, $\text{PANF}(B_A^*) = \text{PANF}(B_A^0) = \mathbf{aANF}$, with no crossing, no inheritance, no cyclicity, no causal production, and no populated successor following from homology alone.

Proved here. The paper adds boundary-status burden, null-route closure, no-crossing, route exhaustion, and kernel minimality to classify the counter-boundary register. The surviving non-degenerate status-class, unique up to governance equivalence, is a fresh internal governed domain satisfying its own kernel and AMetric licensing profile.

Exported by the series. The export of the whole arc is the conditional AASC result: a certified unresolved black-hole endpoint does not transmit parent contents, laws, metricity, energy, information, records, parameters, or causal chains; it forces fresh governed-domain status by constraint recurrence.

Not claimed here. The theorem does not derive a tunneling mechanism, baby-universe dynamics, inherited law, copied constants, energy transfer, information transfer, parent-to-child causation, populated cosmography, biological content, intelligence, technology, or empirical abundance.

Series dependency lock. This capstone starts from the completed homology package and nowhere uses its own conclusion to justify earlier steps. The imported object is

$$B_A^* \neq B_A^0, \quad \text{PANF}(B_A^*) = \text{PANF}(B_A^0) = \mathbf{aANF},$$

together with no-crossing and no-inheritance. The new work here is the counter-boundary classification: once boundary-only terminality, admitted nullity, mute-null-as-classification, inherited continuation, selector/probability, metric/temporal/causal bridge, repair, and richer-scope substitutions are closed, the unique non-degenerate status-class is fresh internal governed-domain status.

Backward-leakage exclusion. The ordinary-language slogan is confined to this paper and means only the theorem-level result just stated. Earlier papers supply antecedents; they do not contain this conclusion. Reading fresh-domain status back into the horizon obstruction, residual-coherence theorem, or homology theorem would reverse the dependency order.

Referee-facing arc firewall. The capstone is not a baby-universe mechanism, tunneling model, causal production story, or inherited-successor cosmology. Its active object is the boundary-pair classification register after homology has already supplied token distinction, equal AMetric role profiles, no-crossing, and no inheritance. The proof therefore cannot use parent energy, parent information, parent law, parent metricity, parent records, parent quantum state, or parent causal order as material for D_1 . No-crossing is not a defect to be repaired; it is the condition that blocks inherited successorhood and forces any non-degenerate survivor to be fresh.

This firewall sets the counterexample burden. A same-scope objection must preserve the boundary-pair target and then exhibit an admissible counter-boundary status outside the closed route register, break a load-bearing imported theorem, show that the antecedents are jointly inconsistent, or prove that boundary-status burden is invalid. A demand for a familiar mechanism, a probability-first selector, a global wavefunction, a common Hilbert space, a remnant archive, a wormhole, a bounce, or an energy-transfer ledger changes scope unless it supplies the exact governance relation it uses. The theorem proves status-existence of a fresh internal governed-domain class, not token selection or populated cosmography.

Pass 6 notation lock. The imported symbol $B_A^\star$ is the endpoint-side New AMetric Boundary token supplied by Paper III, with no-crossing and normal-form homology already fixed. The active object here is the boundary-pair classification register $\mathcal{B}_{CB}$, not a transport domain from D_0 to D_1 . A representative D_1 denotes any token-distinct occupant of the unique $\equiv_{\text{gov}, CB}$ -class; it is not produced by D_0 and is not a successor carrying parent contents. The token $B_A^{D_1}$ denotes the fresh domain's own licensing boundary and remains distinct from both B_A^0 and $B_A^\star$. The predicate FreshGov_{CB} names the survivor governance profile after route exhaustion; it is not an assumed populated cosmology.

Certificate terminology lock. The word *certificate* has four arc-level uses, and the uses are not interchangeable. A *standing-transport certificate* is a same-scope map or witness assignment that preserves a threatened reference class in the exterior reconstruction problem. An *exterior witness certificate* is a concrete exterior, boundary, radiation, remnant, holographic, island, or quotient map that removes a burden from the unresolved-residue ledger. An *endpoint-certified target* is not a rescue certificate; it is the antecedent package in which a nonempty unresolved residue, old-scope standing loss, residual metric-support exhaustion, closed false endpoints, null-route closure, and no prior re-governance make endpoint AMetric exposure the target. A *capstone-certified endpoint* is the Paper IV input: the homology handoff together with endpoint certification and no-crossing. A horizon can be transport-certified and thereby exit the exterior obstruction; an unresolved residue can be endpoint-certified precisely because no such transport certificate has resolved it.

Exterior-quotient and metric-regime lock. The proof-level object of the arc is the declared exterior standing quotient of the black-hole model, not a particular metric-regime representative. In the familiar settled Kerr–Newman example the quotient is displayed by mass, electric charge, and angular momentum, but that example does not restrict the theorem to stationary representatives and does not make absence of hair a separate proof premise. Hair-like, dynamical, evaporating, ringdown, perturbative, semiclassical, quasi-local, high-curvature, or quantum-gravity structure has same-scope force only when it is declared as exterior quotient content, an exterior witness, a leakage channel, a certificate, a non-sink quotient, or richer governance. Such structure subtracts from the unresolved residue exactly to the extent that it re-fixes standing-bearing content. If it does not re-fix standing, it is skin, bookkeeping, hidden load, repair, selector import, carrier laundering, or scope change; it does not provide an additional endpoint route.

Exterior-quotient antecedent lock. The capstone does not use absence of hair as a separate metaphysical premise and does not require stationarity. It imports the endpoint handoff: a nonempty unresolved residue R_H remains after the declared exterior quotient and every standing-bearing certificate have subtracted exactly what they genuinely re-fix. In the classical settled example the exterior quotient is displayed by mass, electric charge, and angular momentum, but richer hair-like or dynamical structures are handled by the same rule. If they re-fix standing, they subtract from R_H ; if they do not, they have no same-scope force against endpoint certification or counter-boundary route closure.

Pass 9 local referee-response lock. For local review, the capstone has to be read in the boundary-pair classification register. It does not ask which parent object crosses into D_1 ; no parent object crosses. It asks which non-degenerate counter-boundary status-class remains once the positive endpoint token B_A^* , no-crossing, null closure, boundary-status burden, and route exhaustion are all fixed. The theorem proves status-existence of a fresh internal governed-domain class up to $\equiv_{\text{gov}, CB}$. It does not select a token cosmology, populate the domain, inherit laws or constants, transfer energy or information, or supply a tunneling/collapse/baby-universe mechanism. Any local objection must either break the boundary-status burden, exhibit a same-scope survivor outside the route register, or show that the capstone antecedents cannot jointly instantiate.

Pass 10 hostile-referee closure lock. The capstone's decisive attack surface is not whether parent contents cross; the theorem agrees that they do not. The decisive attack surface is whether, after a positive endpoint-status token, no-crossing, admitted-null closure, boundary-status burden, and route exhaustion are fixed, there remains a non-degenerate same-scope counter-boundary status outside fresh internal governed-domain status. A successful hostile objection must therefore either break the boundary-status burden, reopen admitted nullity or mute terminality as a classificatory survivor, supply a same-scope survivor outside the route register, or show the antecedents cannot jointly instantiate. Absent that, no-crossing does not silence the theorem; it removes inheritance and leaves only fresh governance as the non-degenerate survivor.

Roadmap and Controlled Claim

The theorem target is the forced status-existence of a fresh internal governed domain under the homologous AMetric licensing condition:

$$\text{certified AASC black-hole endpoint} \Rightarrow \text{fresh constraint-forced universe.}$$

Here *universe* means a fresh internal governed domain under the AASC kernel and AMetric normal form. It does not mean a transported population, inherited spacetime, crossed law package, copied information state, parent energy transfer, causal offspring, or token-selected cosmology.

Why no-crossing does not imply silence

No-crossing blocks parent-scope transport; it does not erase the classification problem. By the time this paper begins, B_A^* is already a positive endpoint-status classification of a nonempty evaluated residue in the boundary-pair register. The active burden is therefore not “what crosses?” but: which counter-boundary status-class remains compatible with this positive endpoint status, no-crossing, null-route closure, and route exhaustion?

Mute absence cannot occupy that register, because mute absence performs no classification. Admitted nullity is a separate positive candidate and is closed by the null-regime theorem. Boundary-only terminality either does no work, collapses into mute or admitted-null classification, or asserts a non-null status and therefore enters the fresh-governance or richer-scope register. Thus no-crossing creates a classification burden rather than a crossing burden: any survivor must be fresh because inheritance is closed, and it must be governed if it is non-degenerate.

No AMetric boundary fusion under mergers

A black-hole merger may unify the exterior metric/source description and produce a single post-merger exterior quotient. That is an exterior AASC-GR fact, not an AMetric boundary operation. Endpoint tokens do not acquire a common metric, adjacency relation, contact predicate, temporal process, causal bridge, operator domain, or cross-boundary standing map merely because the exterior merger has a single ringdown or final parameter ledger.

Thus two New AMetric Boundary tokens cannot collide, touch, fuse, join, or compose. They can be co-classified by an exterior bookkeeping register, for example as a finite family of endpoint statuses, but that family is not a container and not a fused boundary. If a post-merger black-hole target later exposes its own endpoint token, that token is a new classification relative to the merged exterior quotient; it is not inherited from or produced by fusion of the earlier endpoint tokens. Realized interiors collide only inside a shared realized-governance domain. AMetric endpoint tokens coexist without crossing.

The guiding equation is:

$$\begin{aligned} &\text{AMetric homology} + \text{no crossing} + \text{null-route closure} \\ &\quad + \text{kernel minimality} + \text{route exhaustion} \\ &\Rightarrow \text{fresh constraint-forced universe.} \end{aligned}$$

The arrow denotes status-existence in the boundary-pair classification register. It does not denote event production, causal formation, tunneling, transport, or populated realization. The positive claim is that once a certified AASC black-hole endpoint has the same AMetric licensing profile as the origin boundary, and once all non-fresh alternatives close, the same minimal constraints that force any origin-licensed governed-interior architecture force a token-distinct fresh internal governed domain.

The paper has seven jobs.

1. It makes the handoff from AMetric Normal-Form Homology explicit: homology gives token distinction plus role-profile equality, not inheritance.
2. It fixes vocabulary so that *universe*, *successor*, *crossing*, *generation*, *inheritance*, *regime occupation*, *status-existence*, and *populated content* cannot be equivocated.
3. It locks the dependency order from kernel necessity, AMetric boundary discipline, null-regime closure, same-scope exhaustion, black-hole endpoint closure, and AMetric homology to the present theorem.
4. It treats no-crossing as the positive engine of the result: because nothing can be inherited, any non-degenerate counter-boundary domain must be fresh.
5. It exhausts every counter-boundary alternative except fresh constraint-governed interiority.
6. It classifies what is forced without crossing at the kernel, realized-interior, minimal metric-projection, Standard-Model-structural, and downstream parameter/record/history levels, distinguishing role-exact recurrence from non-inherited concrete contents.

7. It proves the constraint-symmetry implication: fresh interiors are role-equivalent by minimal non-degeneracy conditions under matched target-class antecedents, while numerical constants, histories, records, law tokens, and parameter values remain non-inherited but internally constrained. Further downstream specializations are recorded only at their audited conditional status.

Theorem-tier lock

Tier	Result	Status
Tier 1	Fresh internal governed domain-status.	Unconditional under A1–A9 plus the completed closure register.
Tier 2	AASC universe-status.	The same object at the universe-governance register: an internal governed domain under kernel and AMetric licensing.
Tier 3	Realized physical interior roles.	Forced when the realized-physical-interior target class is fixed.
Tier 4	Standard-physics structural envelope.	Forced when the realized-standard-physics target class is fixed.
Tier 5	Numerical constants, empirical records, concrete histories, law tokens, parameter values, and populated contents.	Not inherited, but not arbitrary; require internal certification under the fresh domain’s own admissibility, standing, reference, quotient, witness, calibration, and no-repair conditions.

Target-class hierarchy lock

Layer	Status in this manuscript
Fresh internal governed domain	Unconditional under certified counter-boundary antecedents.
Realized physical interior	Conditional on realized-physical target fixation.
Einstein metric projection	Conditional on minimal Lorentzian first-response target fixation.
Standard-physics structural envelope	Conditional on realized-standard-physics target fixation.
Further downstream specializations	Conditional only under their own audited antecedents. A fixed-domain life-carrier target is not part of the black-hole theorem and is treated later only as a further conditional downstream specialization under its own fixed-domain antecedents.
Intelligence, technology, and detectability	Not part of the black-hole theorem; only downstream conditional status if separately targeted by standing-margin or interface-budget antecedents.

Target-class fixation is an audited theorem condition, not a free stipulation. A target class has standing only when its antecedents, carriers, comparison class, failure modes, and same-scope alternatives are fixed; otherwise it is a label rather than a recurrence theorem.

No-crossing interpretation lock

No parent-scope standing, law, metricity, energy ledger, information package, quantum state, clock, causal chain, memory, record, field, particle identity, operator domain, reference identity, parameter table, or empirical history crosses the New AMetric Boundary. This fixes the universe result as fresh constraint recurrence rather than inheritance. The new domain is role-equivalent at the minimal governance level because the kernel and AMetric normal form are regime-invariant necessities; it is not token-identical, content-identical, law-token-identical, or historically continuous.

This does not imply that downstream contents are arbitrary. It implies that downstream contents cannot be parent-supplied. If numerical constants, empirical records, histories, law tokens, calibration values, or concrete parameter assignments obtain in the fresh domain, they must obtain by internal certification within that domain's own standing-preserving architecture.

Core distinction

Incorrect reading	Theorem reading
No populated successor follows from parent endpoint analysis.	True for the homology theorem alone. The present theorem adds null closure, no-crossing, route exhaustion, and kernel minimality; the result is fresh internal governed domain-status, not inherited population.
Nothing crosses.	Correct. Nothing crosses, and because nothing crosses, the new domain is not inherited. It is fresh.
No law, metric, field, information, or energy is transmitted.	Correct. Therefore role equivalence cannot be genealogical. It is constraint identity: the same minimal conditions force the same role architecture wherever non-degenerate governed domainhood obtains.
The endpoint is not a baby-universe model.	Correct in the transmissive/mechanistic sense. It is stronger in the AASC sense: a certified endpoint forces a fresh constraint regime, i.e. a universe as a new internal governed domain.
The new universe is identical to the old.	Role-equivalent at the minimal governance level, not token-identical, content-identical, parameter-identical, or historically continuous.
Since no parameters, histories, or records cross, downstream structure is unconstrained.	False. Non-crossing blocks parent supply; it does not remove internal admissibility burdens. Any downstream content must be freshly licensed by the new domain's own non-degeneracy conditions.

A fresh universe may have radically different basic structure.	Only by changing target class or declaring richer-scope load. If it is a realized physical interior, the same minimal role package recurs up to admissible equivalence.
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One-page claim-status lock

Misreading	Locked reading
The boundary transmits governance.	It does not. Governance is not transmitted. Fresh internal governed-domain status is forced because inherited, un-governed, null, selector-based, probabilistic, metric, temporal, causal, repaired, and multiple-interior alternatives close.
The theorem proves a baby-universe mechanism.	It does not. It proves a constraint-forced universe in the AASC sense: a fresh internal governed domain, not a tunneling process or parent-child physical production mechanism.
The theorem contradicts no populated successor.	It does not. It denies inherited populated successorhood and proves status-existence of a fresh non-inherited internal governed domain.
Quantum probability can select the far side.	It cannot be boundary authority. Probability is licensed only inside a governed domain; it cannot license the domain whose admissibility it presupposes.
Nullity remains available.	Admitted nullity is closed by the null-regime theorem. Mute nullity has no classificatory force.
The theorem derives parent law continuation.	It denies inherited law. Identical role architecture follows from minimal constraint recurrence, not from crossed law tokens.
Informal slogan.	The admissible formal reading is: certified AASC black-hole endpoints force fresh internal governed domains by constraint recurrence. The transmissive reading is rejected: no parent contents cross.

Handoff from AMetric Normal-Form Homology

The AMetric Normal-Form Homology paper ends at the boundary-profile result:

$$B_A^* \neq B_A^0, \quad \text{PANF}(B_A^*) = \text{PANF}(B_A^0) = \text{aANF}, \quad B_A^* \cong_{\text{ANF}} B_A^0.$$

That theorem deliberately stops short of successor cosmology. Homology alone gives token-distinct equality of complete AMetric role profiles. It gives no crossing map, no common time, no inherited law, no parent energy transfer, no information transport, no causal production, and no populated successor.

The present theorem begins exactly where that theorem stops. It accepts every no-inheritance restriction. It then asks a new classification question: once the homologous endpoint boundary is fixed, once no-crossing is preserved, once admitted nullity is closed, and once non-fresh alternatives are exhausted, can the boundary-pair register remain empty, terminal-only, mute, inherited, ungoverned, probabilistic, temporal, metric, causal, repaired, or multiply occupied? The answer is no.

The handoff is therefore:

Homology paper: $B_A^\star \neq B_A^0$, $\text{PANF}(B_A^\star) = \text{PANF}(B_A^0) = \mathbf{aANF}$, no inheritance.

$\Downarrow$

Present paper: no inheritance + null closure + route exhaustion + kernel minimality
 $\Rightarrow$ fresh internal governed domain.

No-inheritance is the theorem's positive condition. It is the reason the new domain is a fresh constraint regime rather than a derivative, copied, transported, or child-token regime. The role architecture recurs because the kernel and AMetric normal form are the minimal conditions on non-degenerate domainhood, not because anything crosses from D_0 to D_1 .

1 AASC-Internal Status and Dependency Lock

This manuscript is an AASC-internal extension theorem. It does not rederive the kernel, the AMetric boundary, null-regime closure, fixed-domain exhaustion, the black-hole horizon-isolation theorem, the standingless-interior theorem, the New AMetric Boundary theorem, or the AMetric homology theorem. It imports those results as binding dependencies and asks what the combined package forces once the conservative no-inheritance posture is retained rather than softened.

Kernel deletion and the capstone target. The capstone does not become easier if a kernel role is denied. Without admissibility there is no boundary-status gate; without standing there is no reuse-stable governed-domain status; without reference there is no determinate counter-boundary target; without irreversibility, repair and inherited-continuation language can be introduced after closure. Thus deleting a kernel component does not create a rival counter-boundary route. It removes the theorem object whose remaining status is being classified.

1.1 Dependency order

The dependency order is one-way:

kernel $\Rightarrow$ admissibility structure $\Rightarrow$ AMetric boundary
 $\Rightarrow$ fixed-domain exhaustion $\Rightarrow$ null-route closure
 $\Rightarrow$ black-hole endpoint $\Rightarrow$ AMetric homology $\Rightarrow$ counter-boundary regime theorem.

No downstream black-hole result authorizes the kernel or the AMetric invariant. The black-hole endpoint exposes a token of the already-fixed AMetric normal form; it does not generate the invariant. The present theorem similarly does not generate the kernel. It applies kernel necessity and route closure to the counter-boundary classification problem.

1.2 Imported theorem ledger

Imported result	Function in this manuscript
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Kernel of admissibility	Supplies $K_D = (\text{Adm}_D, \text{Stand}_D, \text{Ref}_D, \text{Irr}_D)$ as forced for non-degenerate, identity-preserving, admissibility-assessable construction.
Structure of admissibility	Supplies AMetricity, bivalence, standing as reuse-stable admissibility, uniqueness of the admissible interior, and conservation of standing.
AMetric boundary theorem	Supplies no-selector, no-parameter, no-probability, no-metric, no-canonical-representative, non-eventful boundary discipline.
Null-regime closure	Closes admitted total vacuity as a same-type admissibility-bearing endpoint. Separates mute nullity from admissible null regime.
Unus Solus Possibilis Est	Supplies uniqueness of admissible interior and blocks multiple same-scope admissible interiors.
Closure by exhaustion / fixed-domain exhaustion	Supplies the proof method: same-scope alternatives either alter closed loci, collapse to bookkeeping, or change scope.
Arrow and Clock papers	Block primitive temporal direction, primitive clocks, and boundary-level time/sequence backflow.
Anchor/Tensor/Skin	Supplies fixed-domain role decomposition and blocks fourth-role smuggling.
Metricity as Licensed Description	Blocks metric backflow and primitive metric authority at boundary.
Realized Physical Interior	Supplies forced role architecture once the target is fixed as a realized physical interior.
Standalone Standard Model Structure	Supplies structural Standard-Model envelope once the standard-physics target class is fixed; numerical parameters remain downstream.
Minimal Life Carriers	Supplies only a further conditional target-class specialization: minimal carrier form under fixed-domain life antecedents, including local repair, retained identity-memory, standing margin, standing localization, and finite standing-fixation interval; it does not prove inherited life, biological substrate, intelligence, technology, or abundance.
Einstein Projection	Supplies the licensed metric projection whose black-hole endpoint authority later exhausts.
Black Holes as Licensed Limit Regimes	Supplies horizon isolation, exterior reconstruction burden, and no-totalization.
Standingless Interiors	Supplies residual coherence and governance deficiency after standing transport failure.
New AMetric Boundary	Supplies endpoint closure of the unresolved residue and absolute no-crossing.
AMetric Normal-Form Homology	Supplies $\text{PANF}(B_A^\star) = \text{PANF}(B_A^0) = \mathbf{a}\text{ANF}$ with token distinction and no inheritance.

1.3 Kernel deletion and the counter-boundary register

The counter-boundary theorem depends on the full AASC kernel because the boundary-pair register is not raw silence. Each component supplies one coordinate required to classify the endpoint without allowing inherited successorhood or repair.

Without **Adm**, there is no classification gate. A proposed counter-boundary status could no longer be distinguished from a selector, probability rule, metric bridge, third-domain relation, bookkeeping label, or richer-scope declaration. Route closure would become an arbitrary list rather than a fixed-domain admissibility test.

Without **Stand**, there is no standing-bearing burden that fails to cross. The no-crossing theorem would still deny transport in words, but it would not say what standing-bearing content, if any, was unavailable for inheritance or recovery.

Without **Ref**, the object of no-crossing is indeterminate. There would be no fixed parent item, residue, boundary-pair target, or counter-boundary status whose inheritance, nullity, repair, or freshness is under evaluation.

Without **Irr**, inherited successorhood and repair reopen. A final-state selection, hidden branch, baby-universe label, late dictionary, or repaired operator domain could be introduced after the endpoint and then counted as if it had preserved the same standing history.

Removing a kernel component therefore does not produce an alternative counter-boundary status. It destroys the register in which the capstone is stated. The theorem's burden is not to move parent content across the boundary, but to classify which non-degenerate counter-boundary status remains once the kernel is preserved, no-crossing is accepted, nullity is closed, and every non-fresh route fails.

1.4 Proof obligation of the present paper

The present manuscript has one new load-bearing task: show that no-crossing plus null-route closure and AMetric homology force status-existence of a fresh internal governed domain rather than leaving the counter-boundary classification empty, arbitrary, inherited, probabilistic, temporal, or ungoverned.

The proof must preserve four restrictions.

1. It may not let parent-scope content cross.
2. It may not treat the boundary as a process, event, time, tunnel, or selector.
3. It may not use nullity as an admissible endpoint after null-route closure.
4. It may not overpromote role-level knowability into parent-derived populated contents.

Lemma 1.1 (Boundary-pair comparison scope). The theorem is evaluated on the boundary-pair classification object

$$\mathcal{B}_{CB} = (D_0, B_A^*, \mathbf{aANF}, \mathcal{R}_{\text{closed}}),$$

where $\mathcal{R}_{\text{closed}}$ records the closed parent-continuation, inheritance, null, selector, probabilistic, temporal, metric, causal, repair, and multiplicity routes. It is not evaluated on a transport domain from D_0 into D_1 . Therefore no step of the proof requires a crossing map. The admissible question is which counter-boundary status remains compatible with no-crossing, null closure, AMetric homology, and route exhaustion.

Construction witness. A transport-domain reading would ask which parent object moves into a successor. That question is rejected by absolute no-crossing. The present theorem instead fixes the

endpoint token B_A^* , its AMetric normal form, and the closed route register, then classifies admissible counter-boundary statuses relative to that boundary-pair object. Parent-side endpoint classification and counter-boundary status classification are therefore not the same as parent-to-successor transport. $\square$

2 Vocabulary Discipline

The theorem is vulnerable unless its target words are disciplined. This section fixes the terms used downstream.

Definition 2.1 (Parent regime). A parent regime D_0 is a fixed governed domain whose old-scope standing, reference, metric projection, source ledgers, laws, records, and continuation relations are evaluated under its own AASC kernel and declared physical scope.

Definition 2.2 (New AMetric Boundary). A New AMetric Boundary B_A^* is the endpoint-side AMetric boundary exposed when the unresolved horizon-isolated residue reaches maximum governance-depth and old-scope metric, temporal, causal, source-ledger, operator, storage, transfer, repair, and null routes are exhausted. It is not a geometric horizon, spatial surface, temporal moment, or dynamical event.

Definition 2.3 (Counter-boundary regime). A counter-boundary regime D_1 , as used in representative notation, denotes a token-distinct member of the fresh internal governed-domain status-class constrained by the homologous AMetric boundary normal form. It is not a successor in the inheritance sense. It is not a continuation of D_0 . It is not caused by D_0 . It is not populated by crossed parent contents.

Definition 2.4 (Regime occupation). Regime occupation is non-null governed domain-status. It means that the counter-boundary side is not admitted as vacuity, not an ungoverned residue, not inherited old-scope continuation, and not a boundary selector. It does not by itself enumerate fields, particles, constants, observers, histories, or records.

Definition 2.5 (Internal governed domain). An internal governed domain is a non-null domain-status that instantiates the AASC kernel under its own admissibility gate, standing relation, reference-fixation discipline, irreversibility/no-repair structure, and AMetric licensing boundary. It is internal because its standing, reference, lawlike roles, comparison structures, and continuation conditions are fixed under its own governance rather than by parent transfer or boundary selection.

Definition 2.6 (AASC universe). An AASC universe is an internal governed domain at the universe-governance register: its governance is sufficient to support interior law, metricity, temporality, causality, records, and physical-role articulation when the realized-physical-interior target class is fixed. Its universe-status is governance-level. It does not require inherited parent contents, parent metrics, parent records, parent energy ledgers, parent information packages, parent law tokens, or parent histories.

Definition 2.7 (Constraint-forced universe). A constraint-forced universe is an AASC universe whose status-existence is forced by kernel necessity and AMetric boundary discipline once non-governed, null, inherited, selector-based, probabilistic, metric, temporal, causal, repaired, and multiple-interior alternatives close. It is not an inherited population, not a transported law package, not a copied metric interior, not a causal product, and not parent-derived cosmography. Its universe-status consists in fresh non-degenerate governed interiority under the minimal constraints required for admissibility, standing, reference, and irreversibility.

Proposition 2.8 (Universe-register equivalence). Within AASC, the universe-level object at the governance register is not a populated token cosmology. It is a non-null internal governed domain whose

kernel and AMetric licensing conditions are sufficient for realized-physical-interior role articulation when that target class is fixed. Therefore proving fresh internal governed domain-status is exactly proving universe-status at the AASC governance register. Additional claims about metric realization, field inventory, standard-physics structure, numerical parameters, histories, records, or occupants remain downstream target-class or certificate matters and are not inherited from the parent regime.

Construction witness. An AASC universe is typed by admissible internal domainhood before it is typed by populated cosmography. If the domain is non-null, internally governed, kernel-instantiating, and AMetric-licensed, it has the governance structure required for universe-status. If the realized-physical-interior target is fixed, the standard physical role ladder may be recovered under its own antecedents. If that target is not fixed, the theorem still proves the governance-register universe object and withholds downstream population, parameter, and history claims. Thus the word universe is not a synonym added after the proof; it names the AASC register occupied by a fresh internal governed domain. $\square$

Universe-register lock. In AASC, universe-status at this theorem level is internal governed-domain status, not inherited populated cosmography. The theorem proves the governance-register universe object. Metric realization, field inventory, Standard-Model structure, numerical parameters, records, histories, and occupants require their own target-class antecedents or internal certificates.

Definition 2.9 (Structural identity without inheritance). Structural identity without inheritance obtains when two token-distinct governed domains instantiate the same role architecture because both satisfy the same minimal non-degeneracy constraints, not because any law, metric, field, particle, source ledger, record, information package, energy ledger, parameter, or history crosses from one to the other. In this sense, the counter-boundary regime is identical to any origin-licensed governed-interior architecture at the level of kernel and AMetric role structure, while remaining token-distinct and non-inherited.

Definition 2.10 (Constraint recurrence). Constraint recurrence is the reappearance of the same minimal governance architecture at a token-distinct boundary condition because the AASC kernel and AMetric normal form are regime-invariant. Constraint recurrence is not cyclicity, copying, transport, reproduction by information transfer, or causal production. It is sameness by necessity: every non-degenerate governed domain must satisfy the same minimal roles.

Definition 2.11 (Populated content). Populated content means concrete internal occupants: particular law tokens, fields, particles, metric realization, dimensionality, numerical parameters, records, histories, observers, or empirical facts. Populated content is not inherited from D_0 under the theorem.

Definition 2.12 (Fresh internal governed-domain status). Fresh internal governed-domain status is kernel instantiation under AMetric boundary discipline without transmission from D_0 . It is not a crossed law package, a parent-scope operator, an inherited gate, a metric bridge, a hidden transport relation, or a crossed source ledger. It denotes governance whose admissibility, standing, reference, and irreversibility are fixed under its own domain typing rather than inherited from the parent regime.

Remark 2.13 (Survivor status is not definitional). That a fresh internal governed domain is the unique non-degenerate counter-boundary status-class up to $\equiv_{\text{gov}, CB}$ is not part of this definition. It is proved below by boundary-status burden, route-register exhaustion, null-route closure, absolute no-crossing, and AMetric homology.

Definition 2.14 (No crossing). No crossing means that no parent-scope standing-bearing item survives as the same item in D_1 . The denied crossing includes standing, reference identity, law package, metricity,

energy/source ledger, information carrier, field, particle identity, clock, causal chain, memory, record, quantum state, and operator domain.

Definition 2.15 (Counter-boundary licensing token). If X is a freshly governed counter-boundary regime, B_A^X denotes its own AMetric licensing boundary token. For the regime denoted D_1 , write $B_A^{D_1}$. This token is not B_A^* , not B_A^0 , and not a crossed parent object. It is the boundary token under which X 's own governance is licensed.

Lemma 2.16 (Boundary-token discipline). The boundary tokens B_A^0 , B_A^* , and $B_A^{D_1}$ are token-distinct even when their complete AMetric normal-form profiles agree:

$$\text{PANF}(B_A^0) = \text{PANF}(B_A^*) = \text{PANF}(B_A^{D_1}) = \text{aANF},$$

while

$$B_A^0 \neq B_A^*, \quad B_A^* \neq B_A^{D_1}, \quad B_A^{D_1} \neq B_A^0.$$

The endpoint token B_A^* does not become or transmit $B_A^{D_1}$. It fixes the boundary-pair classification condition exposed by old-scope exhaustion. A representative D_1 has its own licensing token $B_A^{D_1}$. The equality above is role-profile equality under minimal AMetric constraint, not token identity, transition, or crossing.

Construction witness. B_A^0 is the origin-side licensing token of the parent interior. B_A^* is the endpoint-side New AMetric Boundary token exposed by parent old-scope exhaustion. $B_A^{D_1}$ is the licensing token of a representative fresh internal governed domain. Identifying any two would either collapse theorem positions, assert transmission, or erase no-crossing. Homology licenses equality of role profiles only; token identity remains excluded. $\square$

Definition 2.17 (Fresh counter-boundary governance). A candidate counter-boundary status X satisfies $\text{FreshGov}_{CB}(X)$ iff all of the following hold:

- FG1. X is non-null and classificatory, not mute absence.
- FG2. X is not inherited from D_0 .
- FG3. $\text{NoCross}(D_0, X)$: no parent-scope item crosses into X as the same standing-bearing item.
- FG4. X instantiates the kernel roles ($\text{Adm}_X, \text{Stand}_X, \text{Ref}_X, \text{Irr}_X$) under its own domain typing.
- FG5. The licensing boundary of X satisfies $\text{PANF}(B_A^X) = \text{aANF}$.
- FG6. Metric, temporal, causal, probabilistic, selector, repair, and operator authority are downstream of X 's own governance and are not boundary-authorizing substitutes.

Remark 2.18 (Predicate discipline). FreshGov_{CB} names the candidate status identified by the theorem once the route register is evaluated. Its survivor status is not definitional. The predicate records the positive governance profile: non-nullity, non-inheritance, no-crossing, kernel instantiation, AMetric licensing, and downstream-only metric, temporal, probabilistic, quantum, causal, selector, repair, and operator authority.

Definition 2.19 (Counter-boundary governance equivalence). Let X and Y be counter-boundary status candidates. Write

$$X \equiv_{\text{gov}, CB} Y$$

iff X and Y agree on all admissibility-bearing governance coordinates:

- GE1. non-null classificatory status;
- GE2. non-inheritance from D_0 ;

- GE3. no-crossing of parent-scope standing, law, metricity, source or energy ledger, causal chain, information carrier, record, or identity;
- GE4. instantiation of the kernel roles $\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}$ at the governance-profile/status-coordinate level, not merely the presence of analogous vocabulary, concrete internal law, token content, or presentation;
- GE5. AMetric licensing profile $\text{PANF}(B_A^X) = \text{PANF}(B_A^Y) = \mathbf{aANF}$;
- GE6. closure status of inherited continuation, nullity, ungoverned residue, selector/probability routes, metric/temporal/causal bridges, and same-scope repair;
- GE7. downstream-only status of metric, temporal, causal, probabilistic, quantum, selector, repair, and operator authority.

They may differ in token presentation, internal realization, numerical parameters, concrete law tokens, field inventory, population, or empirical history without differing as counter-boundary governance classes.

Remark 2.20 (Uniqueness target). The uniqueness theorem below concerns $\equiv_{\text{gov}, CB}$ -classes. It does not assert numerical identity of one token universe, a unique concrete cosmology, physical-token production, populated realization, or a preferred internal history.

Lemma 2.21 ($\equiv_{\text{gov}, CB}$ preserves all standing-relevant coordinates). The equivalence relation $\equiv_{\text{gov}, CB}$ does not obtain uniqueness by ignoring admissibility-bearing differences. It quotients only token presentation, internal realization, numerical parameter values, concrete law tokens, field inventories, populations, and empirical histories. It preserves every coordinate relevant to the theorem target: non-null classificatory status, non-inheritance, no-crossing, kernel instantiation, AMetric licensing, closure of non-fresh boundary-authority routes, and downstream-only status of metric, temporal, causal, probabilistic, quantum, selector, repair, and operator authority.

Construction witness. The counter-boundary theorem classifies governance status, not populated content. Any difference in admissibility gate, standing relation, reference-fixation role, irreversibility, AMetric licensing, no-crossing, inherited continuation, nullity, selector authority, metric bridge, temporal sequence, causal production, repair, or boundary operator changes a coordinate preserved by $\equiv_{\text{gov}, CB}$. Differences in token realization, parameter values, law tokens, histories, populations, or internal cosmography are downstream of the governance status and are not part of the equivalence target. Therefore uniqueness up to $\equiv_{\text{gov}, CB}$ is uniqueness over exactly the admissibility-bearing coordinates fixed by the theorem. $\square$

Definition 2.22 (Status-existence and populated-existence). A counter-boundary status has *status-existence* when it is forced as a typed admissibility classification within the boundary-pair register. Status-existence is existence as a governance-status object.

A counter-boundary regime has *populated-existence* only when concrete internal occupants, law tokens, fields, metric realization, histories, records, or empirical contents are independently licensed inside that regime.

The main theorem proves status-existence of the fresh internal governed-domain class. It does not derive populated-existence, physical-token production, empirical realization, or internal cosmography.

Remark 2.23 (Representative notation). When the manuscript writes D_1 , it denotes an arbitrary representative of the unique $\equiv_{\text{gov}, CB}$ -class. The notation does not assert that AASC selects a unique concrete token, a unique cosmology, or a populated physical universe.

2.1 Forbidden synonyms

The paper avoids the word *successor* except when rejecting inherited successorhood. It avoids *after* except as external-use language. It avoids *produces*, *forms*, *emerges from*, *transmits*, *continues into*, and *branches from* unless those are explicitly classified as inadmissible parent-scope descriptions.

3 The Parent-Scope Endpoint Result

This section restates the parent-side black-hole sequence only to fix the theorem object. The sequence has four layers.

3.1 Horizon isolation and the burden

A black-hole target is fixed by its exterior reconstruction scope. The old exterior domain has an exterior standing quotient, witness algebra, continuation relation, parameter ledger, thermodynamic ledger, and certificate status. Horizon isolation threatens reference-standing classes when later exterior data cannot re-fix the same classes through the old witness system.

Let B_H be the horizon-generated reference-standing burden. Let $W_{\text{ext}}(B_H)$ be the part of that burden resolved by genuine standing-bearing exterior witness relations. Define the unresolved horizon-isolated residue by

$$R_H = B_H \setminus W_{\text{ext}}(B_H).$$

This is role-ledger notation, not a primitive set of information tokens.

3.2 Residual coherence and governance deficiency

When old-scope standing transport fails, the horizon-isolated carrier is not thereby null. It may retain residual coherence: role-indexed organization inherited from the pre-break regime without full standing. Residual coherence is not standing, not ordinary interior matter, not measurement standing, not energy-ledger continuity, and not a new AMetric reset at the horizon.

The horizon is a derivative metric boundary inside the licensed parent projection. It is not the origin-side AMetric boundary and not the New AMetric Boundary. The New AMetric Boundary appears only at maximum governance-depth, after residual support for the licensed metric projection exhausts.

3.3 Endpoint route closure

At maximum governance-depth the unresolved residue cannot be classified as:

- a classical singularity, because singularity is a metric diagnosis internal to a still-operative metric articulation;
- a remnant or archive, because storage requires standing-bearing carrier identity and continuation;
- an information sink, because information is not primitive cargo and requires carrier, encoding, reference, witness, quotient, and standing;
- a bounce or transition, because transition requires time, causality, metric structure, and operator domain;

- a wormhole or throat, because passage requires old-scope metric/causal continuation;
- an energy-transfer site, because energy-transfer language requires a source ledger, clock, and cross-boundary comparison relation;
- admitted nullity, because total vacuity cannot carry admissibility-bearing endpoint status.

3.4 The endpoint conclusion

When $R_H \neq \emptyset$ and residual metric successor support is exhausted at maximum governance-depth, the endpoint closure-status of R_H is B_A^* , a New AMetric Boundary. It is structurally homologous to the origin-side AMetric boundary:

$$B_A^* \neq B_A^0, \quad \text{PANF}(B_A^*) = \text{PANF}(B_A^0) = \text{aANF}.$$

The equality is role-profile equality, not token identity, transition, inheritance, or cyclic return.

4 Absolute No-Crossing

No-crossing is the positive condition that prevents inherited successorhood and thereby forces any non-degenerate counter-boundary domain to be fresh. Once nullity, boundary-only terminality, and degenerate alternatives are closed, no-crossing becomes the engine of constraint recurrence rather than a disclaimer.

Theorem 4.1 (Absolute no-crossing). Let B_A^* be a New AMetric Boundary exposed at maximum governance-depth for a nonempty unresolved horizon-isolated residue R_H . Then no parent-scope standing-bearing item has a same-item image across B_A^* . In particular, no parent standing, reference identity, law, metric, source ledger, energy ledger, information package, field, particle identity, quantum state, clock, causal chain, memory, record, branch label, or operator domain crosses B_A^* .

Construction witness. A same-item image across B_A^* would require a cross-boundary transport rule preserving target identity, standing, quotient identity, and lawful continuation. The AMetric boundary admits no metric, temporal, causal, probabilistic, selector, operator, canonical representative, inherited law, or transport rule with boundary authority. If the rule is imported from D_0 , it violates no-crossing and old-scope exhaustion. If it is supplied by the boundary, it violates AMetric non-transmissivity. If it is supplied from a third domain, it is richer-scope governance and not a same-scope crossing relation. Thus no parent-scope item crosses as the same standing-bearing item. $\square$

Corollary 4.2 (No inherited successor). There is no inherited successor of D_0 across B_A^* . Any non-degenerate representative of the counter-boundary class is token-distinct and freshly governed.

Corollary 4.3 (No destruction event). No-crossing is not information destruction, energy destruction, matter destruction, or law destruction. Destruction is an event classification internal to a temporal, causal, metric, and ledger-bearing regime. Those cross-boundary conditions are absent.

Corollary 4.4 (No creation event). No-crossing is likewise not creation from parent material or creation from nothing. Creation is also a temporal-causal event classification. The theorem classifies counter-boundary governance status after endpoint route exhaustion; it does not supply a parent-causal production event.

4.1 No-crossing ledger

Parent-scope item	Why crossing is blocked	What remains possible
Standing	Requires same-scope admissible continuation and target identity.	Fresh standing under D_1 's own gate.
Reference identity	Requires a cross-boundary target-fixation map.	New reference roles inside the fresh internal governed domain.
Law package	Requires inherited admissibility authority.	Law-like roles inside D_1 if realized.
Metric	Requires metric comparison at the boundary.	Metricity licensed downstream in D_1 .
Energy/source ledger	Requires common ledger, clock, and comparison carrier.	Source roles inside D_1 , not parent energy transfer.
Information	Requires carrier, encoding, witness, quotient, and reuse.	New information roles if D_1 realizes record carriers.
Quantum state	Requires Hilbert/predictive structure and boundary transport.	Quantum structure only as downstream governed interior.
Clock/time	Requires clock quotient or temporal-order role.	Time/clock roles only after fresh interior licensing.
Causality	Requires witnessed dependence across an admitted order.	Causality only as interior dependence structure.
Memory/record	Requires retained witness identity and standing.	New records only inside D_1 .

5 Route Exhaustion Beyond the Boundary

The survivor-by-exhaustion step is valid only if the alternative register is complete. AASC supplies completeness through fixed-domain exhaustion: a same-scope alternative must change boundary typing, evaluated classification, quotient identity, continuation structure, role carrier, or closed operator status. If it changes none, it is bookkeeping. If it changes one, it is classified by the route matrix below.

5.1 Alternative register

Candidate counter-boundary status	Failure mode	Disposition
Same-regime continuation	Requires old-scope standing transport across B_A^* .	Closed
Inherited governance	Requires parent law/gate/metric/ledger authority to cross.	Closed
Ungoverned residue	Lacks admissibility, standing, reference, irreversibility, and continuation.	Closed
Admitted nullity	Total vacuity cannot be an admissibility-bearing regime.	Closed

Mute nullity		If mute, it has no classificatory force; if used to govern, it ceases to be mute.	Inert/closed
Boundary-only terminality		If it does no counter-boundary classificatory work, it is not a counter-boundary candidate; if it denies all non-null status while functioning as a status claim, it collapses into admitted nullity or mute-null-as-classification; if it asserts non-null status, it enters fresh internal governed-domain or richer-scope classification.	Classified
Probabilistic selection		Probability requires admitted state space, measure, alternatives, and standing; it is interior, not boundary authority.	Closed
Quantum uncertainty at boundary		Quantum structure is downstream of governance and cannot license governance.	Closed
Primitive time sequence		Time and clock roles are licensed interior quotient roles.	Closed
Metric or causal bridge		Metricity and causality are licensed descriptions inside governed regimes.	Closed
Same-scope repair/totalization	re-	Operator closure blocks same-scope repair, totalization, and deferred admissibility.	Closed
Boundary selector or canonical representative		AMetric no-selector/no-parameter theorem blocks this.	Closed
Multiple rival interiors		Unique admissible interior blocks same-scope multiplicity; token plurality requires distinct governance.	Closed
Fresh but non-universe governed status		If it is non-null, internal, governed, and kernel-instantiating, it already satisfies the AASC universe criterion at the minimal governance level; if not internal or not governed, it collapses into skin, mute nullity, ungoverned residue, or richer-scope declaration.	Classified
Skin-only external description		Changes no standing-bearing status.	Bookkeeping
Richer third-domain theory		Not same-scope; must declare its own governance and cannot count as crossing.	Scope change
Fresh constraint-forced universe	uni-	Instantiates kernel under AMetric normal form without crossing; realizes fresh internal governed domain-status.	Survives

5.2 Fixed-domain locus matrix

The route register is not an informal list. It is the image of the fixed-domain comparison loci under counter-boundary classification.

Fixed-domain locus	Candidate forms	Closure source
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Boundary typing	Primitive metric, time, probability, selector, canonical representative, inherited law, or fresh AMetric licensing.	AMetric boundary theorem; no-selector/no-parameter closure.
Evaluated classification	Admitted nullity, mute nullity, boundary-only terminality, ungoverned residue, or fresh internal governed domain-status.	Null-regime theorem; kernel necessity; bivalence; boundary-status burden.
Quotient/reference identity	Inherited information, memory, law, record, standing, or reference identity.	Absolute no-crossing; reference failure; standing conservation.
Continuation/transport	Same-regime continuation, inherited governance, metric bridge, causal bridge, temporal sequence, or third-domain relation.	Horizon no-totalization; metricity, clock, and standing-arrow discipline.
Role carrier	Skin-only description, richer-scope declaration, multiple-interior claim, or role recurrence.	Anchor/tensor/skin; Unus; scope discipline.
Operator/repair status	Same-scope totalization, repair, probability-first route, or quantum-first crossing.	Same-scope operator closure; no-repair; AMetric no-boundary-authority discipline.

Lemma 5.1 (Route-register exhaustiveness). Every admissibility-relevant counter-boundary candidate must enter through one of the fixed-domain loci: boundary typing, evaluated classification, quotient/reference identity, continuation or transport structure, role carrier, or operator/repair status. The locus matrix and the alternative register exhaust those loci.

Construction witness. Fixed-domain exhaustion permits no admissibility-relevant variation outside the six loci in the matrix. A candidate affecting boundary typing appears as primitive metric, time, probability, selector, canonical representative, inherited law, or fresh AMetric licensing. A candidate affecting evaluated classification appears as admitted nullity, mute nullity used as classification, boundary-only terminality, ungoverned residue, or fresh internal governed-domain status. Boundary-only terminality does not define a seventh locus: if it does no counter-boundary classificatory work, it is not a counter-boundary candidate; if it denies all non-null status while functioning as a status claim, it collapses into admitted nullity or mute-null-as-classification; if it asserts non-null status, it enters the fresh internal governed-domain or richer-scope register. A candidate affecting quotient or reference identity appears as inherited reference, information, memory, record, law, or standing. A candidate affecting continuation or transport appears as same-regime continuation, inherited governance, metric bridge, causal bridge, temporal sequence, or richer third-domain relation. A candidate affecting role carrier appears as skin-only description, declared richer scope, multiple interior claim, or target-class role recurrence. A candidate affecting operator or repair status appears as same-scope totalization, repair, probabilistic selection, or quantum-first crossing. If a proposal affects none of these loci, it changes no standing-bearing classification and is bookkeeping. If it affects one, it falls under the route table. No additional same-scope locus remains. $\square$

Corollary 5.2 (No fifth counter-boundary route). Inside the boundary-pair comparison scope, there is no counter-boundary route outside the exhausted register. A putative fifth route must either reduce to a listed route, change scope, or perform no standing-bearing work.

Lemma 5.3 (Boundary-status burden). Once $R_H \neq \emptyset$ reaches maximum governance-depth and is classified as B_A^* , the result is not raw silence or absence of classification. It is a positive endpoint-status result for the unresolved residue. Therefore the boundary-pair object carries a status burden: every admissibility-relevant counter-boundary candidate must be typed in the fixed-domain register as inherited continuation, admitted nullity, mute non-classification, boundary-only terminality, ungoverned residue, selector/probability/metric/temporal/causal repair route, richer-scope declaration, skin, or fresh internal governed-domain status. The burden is not an existential postulate; it is the fixed-domain obligation created by a nonempty evaluated residue already classified as an endpoint.

Construction witness. The burden follows from three upstream locks. First, $R_H \neq \emptyset$ places a nonempty residue in the evaluated endpoint fragment. Second, classification as B_A^* is a positive endpoint-status result, not raw silence. Third, fixed-domain bivalence and completion require every admissibility-relevant evaluated residue to receive a typed status in the declared register: admitted continuation, inadmissible route, null claim, mute non-classification, boundary-only terminality, scope change, bookkeeping, skin, richer-scope declaration, or fresh internal governed-domain status. Mute absence may remain mute only by doing no classificatory work; it cannot both preserve the positive endpoint-status result and terminate the status burden. Admitted nullity is a separate candidate and is closed by the null-regime theorem. What remains is the boundary-pair status burden recorded in the route register. $\square$

Lemma 5.4 (Counter-boundary occupation necessity). Given B_A^* , the boundary-status burden, absolute no-crossing, null-route closure, and AMetric homology, the boundary-pair classification cannot terminate in mute absence or boundary-only terminality while still preserving endpoint-status and avoiding null closure. Every admissible counter-boundary status is either one of the closed routes in the route register or satisfies FreshGov_{CB} .

Construction witness. By the Boundary-status burden lemma, the New AMetric Boundary is already a positive endpoint-status classification of the unresolved residue; it is not raw silence. Admitted nullity is closed by the null-regime theorem. Mute nullity cannot classify, deny, exempt, or occupy anything; if it is used to do so, it ceases to be mute and incurs the kernel obligations. Boundary-only terminality likewise supplies no independent survivor: if it does no counter-boundary work, it is inert; if it denies all non-null status while claiming status authority, it is nullity or mute-null-as-classification; if it asserts non-null status, it enters the fresh internal governed-domain or richer-scope register. Ungoverned residue lacks admissibility, standing, reference, and irreversibility. Inherited routes violate no-crossing. Selector, probabilistic, temporal, metric, causal, quantum, and repair routes are closed by AMetric boundary discipline and operator exhaustion. By route-register exhaustiveness, no further non-fresh locus remains. Therefore any admissible non-degenerate counter-boundary status-class must be fresh internal governed domain-status. $\square$

Theorem 5.5 (Homologous Boundary Licensing Constraint). AMetric normal-form homology does not transmit contents, laws, metricity, source ledgers, standing, or reference identity. However, once admitted nullity, inherited continuation, ungoverned residue, selector/probability routes, metric/temporal/causal bridges, and same-scope repair are closed, any non-degenerate counter-boundary status-class compatible with $\text{PANF}(B_A^*) = \text{PANF}(B_A^0) = \mathbf{aANF}$ must be licensed under the AMetric normal form and must instantiate a fresh internal governed domain rather than inherited governance.

Construction witness. Homology supplies role-profile equality, not transfer. No-crossing blocks inherited governance. Null-route closure blocks admissible vacuity. Kernel necessity blocks ungoverned occupation. AMetric boundary discipline blocks selector, probability, metric, temporal, causal, and operator authority at the boundary. Therefore any non-degenerate status-class compatible with the homologous

boundary profile must be AMetric-licensed under its own domain typing and must instantiate a fresh internal governed domain rather than inherited governance. $\square$

Lemma 5.6 (Minimal Constraint Identity without Inheritance). Let X be any non-degenerate governed domain in the counter-boundary comparison scope. Then X must instantiate the AASC kernel roles ($\text{Adm}_X, \text{Stand}_X, \text{Ref}_X, \text{Irr}_X$) and must be licensed under an AMetric boundary profile. If $\text{PANF}(B_A^\star) = \text{PANF}(B_A^0) = \text{aANF}$, then any fresh non-degenerate counter-boundary representative is role-equivalent to any origin-licensed governed-interior architecture at the level of minimal governance roles. This identity is not inherited. It follows from the minimality and regime-invariance of the kernel and AMetric normal form.

Construction witness. Non-degenerate domainhood is not a bare presentation. It must distinguish licensed from unlicensed use, fix targets, stabilize standing under reuse, and prevent failed acts from becoming the same successful acts by later repair. These are exactly the kernel roles. AMetric boundary discipline blocks primitive metric, temporal, causal, probabilistic, selector, operator, and inherited-law authority at the boundary. Homology fixes the endpoint-side boundary profile to the same complete role vector as the origin-side licensing boundary. Therefore the recurrence is structural: both boundary positions instantiate the same minimal role conditions. Since no parent-scope item crosses, the identity cannot be genealogical or transmissive. It is identity by minimal constraint. $\square$

Theorem 5.7 (Constraint Identity from Homologous Licensing). Assume $\text{PANF}(B_A^\star) = \text{PANF}(B_A^0) = \text{aANF}$, absolute no-crossing, null-route closure, and closure of inherited, ungoverned, selector, probability, temporal, metric, causal, repair, and multiplicity routes. Then any admissible non-degenerate counter-boundary domain is a fresh constraint-forced universe: a fresh internal governed domain role-equivalent to any origin-licensed governed-interior architecture at the level of kernel and AMetric role structure, and non-identical at the level of token, population, history, law token, metric token, parameter table, and empirical inventory.

Construction witness. The closed routes eliminate every non-fresh candidate: inherited continuation violates no-crossing; nullity violates null-route closure; ungoverned residue lacks the kernel; probability, time, metricity, causality, quantum state, selector, canonical representative, and repair presuppose downstream governance or violate AMetric boundary discipline; multiple same-scope interiors violate uniqueness. The Minimal Constraint Identity Lemma then supplies the positive content of the survivor: any remaining non-degenerate domain must instantiate the same kernel and AMetric role architecture as any admissible interior. Because no parent content crosses, the survivor is fresh. Because the boundary profile is homologous with the origin profile, the fresh domain is structurally a universe in the AASC sense. $\square$

6 The Main Theorem

6.1 Formal antecedents

Let the following antecedents be fixed.

Exterior-quotient antecedent. Before these antecedents are evaluated, the declared exterior quotient and all certificate-bearing structures have already subtracted what they genuinely re-fix. Hair-like, dynamical, semiclassical, perturbative, ringdown, boundary, remnant, non-sink, holographic,

island, or richer-scope structures are not ignored. They either remove standing-bearing burdens from R_H , declare a new target, or have no same-scope force. The capstone begins only with the nonempty unresolved remainder that survives that subtraction.

- A1. D_0 is a certified parent regime with kernel K_{D_0} and declared black-hole exterior reconstruction scope.
- A2. Horizon isolation fixes a reference-standing burden B_H .
- A3. Genuine exterior witness relations resolve $W_{\text{ext}}(B_H)$, leaving $R_H = B_H \setminus W_{\text{ext}}(B_H)$.
- A4. $R_H \neq \emptyset$.
- A5. R_H retains role-indexed residual coherence but no same-scope standing transport.
- A6. Residual metric successor support exhausts at maximum governance-depth.
- A7. The endpoint status of R_H is the New AMetric Boundary B_A^* .
- A8. $\text{PANF}(B_A^*) = \text{PANF}(B_A^0) = \text{aANF}$, with $B_A^* \neq B_A^0$.
- A9. Absolute no-crossing holds at B_A^* .

Lemma 6.1 (Completed closure register). Under A1–A9 and the route-register analysis above, the inherited-continuation, boundary-only terminality, admitted-null, mute-null-as-classification, ungoverned-residue, selector/probability, metric/temporal/causal bridge, repair/totalization, skin-authority, storage, transfer, and undeclared richer-scope routes are closed by the cited dependency theorems or local lemmas. This lemma records the completed route-register output; it does not assume the fresh internal governed-domain survivor.

Construction witness. Each listed route appears in the alternative register and in the fixed-domain locus matrix. Inherited continuation and inherited governance violate no-crossing. Boundary-only terminality is inert when it does no counter-boundary work, null-closed when it denies all non-null status while claiming status authority, and classified when it asserts non-null counter-boundary status. Admitted nullity is closed by the null-regime theorem; mute-null-as-classification either remains non-classificatory or incurs non-null governance obligations. Ungoverned residue lacks the kernel. Selector, probability, metric, temporal, causal, quantum, representative, repair, and totalization routes are closed by AMetric boundary discipline and same-scope operator closure. Skin-authority is bookkeeping; storage and transfer require crossed carriers or ledgers; undeclared richer scope is not same-scope. Thus the completed closure register is the output of route analysis, not a premise selecting fresh internal governed-domain status. $\square$

Theorem 6.2 (Counter-Boundary Universe Theorem). Under A1–A9 and the Completed Closure Register Lemma, the boundary-pair classification register contains exactly one admissible counter-boundary status-class up to $\equiv_{\text{gov}, CB}$. That class has status-existence and is represented by any token-distinct D_1 satisfying $\text{FreshGov}_{CB}(D_1)$:

$$K_{D_1} = (\text{Adm}_{D_1}, \text{Stand}_{D_1}, \text{Ref}_{D_1}, \text{lrr}_{D_1}), \quad \text{PANF}(B_A^{D_1}) = \text{aANF}.$$

Equivalently, the survivor is a fresh internal governed domain: a constraint-forced universe in the AASC sense. Its identity with any origin-licensed governed-interior architecture is role-architectural, not genealogical. No parent-scope content, law, standing, metric, energy/source ledger, information carrier, quantum state, clock, causal chain, memory, record, field, particle identity, operator domain, parameter value, or reference identity crosses from D_0 into any representative D_1 . The theorem proves status-existence of the governance-equivalence class and its forced minimal role architecture. It does not derive populated-existence, token selection, inherited populated content, or parent-derived internal cosmography.

Proof. The boundary-pair comparison scope fixes $(D_0, B_A^*, \mathbf{aANF}, \mathcal{R}_{\text{closed}})$, not a transport domain from D_0 to D_1 . The Boundary-status burden lemma prevents the classification from terminating in mute absence while preserving B_A^* as endpoint-status. By A7 and A8, B_A^* is a token-distinct AMetric boundary whose complete normal-form profile equals the origin-side profile. By A9, no parent-scope standing-bearing item crosses B_A^* . Same-regime continuation and inherited governance are therefore unavailable.

By route-register exhaustiveness, every admissibility-relevant counter-boundary candidate enters through boundary typing, evaluated classification, quotient/reference identity, continuation/transport, role carrier, or operator/repair status. By the null-regime theorem and the Completed Closure Register Lemma, admitted vacuity and boundary-only terminality cannot be the counter-boundary status. Mute nullity carries no classificatory force, and ungoverned residue lacks the kernel roles required for non-degenerate occupation. Probability, quantum uncertainty, primitive time, metric passage, causal bridge, selector use, canonical representative selection, and same-scope repair are downstream or prohibited structures; each either presupposes governance, violates AMetric boundary discipline, or exits the comparison class. Multiple rival same-scope interiors are closed by uniqueness of the admissible interior. A richer third-domain construction may be declared, but then it is independent governance rather than parent-scope crossing.

The Homologous Boundary Licensing Constraint supplies the non-transmissive licensing force: any non-degenerate counter-boundary status-class compatible with $\text{PANF}(B_A^*) = \text{PANF}(B_A^0) = \mathbf{aANF}$ must be AMetric-licensed and fresh rather than inherited. By counter-boundary status necessity, the remaining admissible non-degenerate status-class must satisfy FreshGov_{CB} . Kernel necessity supplies $(\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr})$ for that class; the AMetric boundary normal form supplies $\text{PANF}(B_A^{D_1}) = \mathbf{aANF}$; no-crossing blocks parent-token inheritance. Thus any D_1 satisfying FreshGov_{CB} represents the unique counter-boundary status-class up to $\equiv_{\text{gov}, CB}$. Since D_1 instantiates the same minimal kernel and AMetric role architecture as any origin-licensed governed-interior architecture, the survivor is a universe in the AASC sense. Since no parent-scope content crosses, that universe is fresh rather than inherited. $\square$

Corollary 6.3 (Black-Holes-Make-Universes, AASC form). Under the theorem's antecedents, a certified AASC black-hole endpoint forces fresh internal governed-domain status in the AASC sense: it forces the status-existence of a fresh internal governed domain under the same minimal kernel and AMetric constraints that license any origin-governed interior. This is not causal production, material transfer, law inheritance, information transport, energy transfer, or populated-token derivation. It is constraint recurrence of non-degenerate governed interiority.

Corollary 6.4 (No inherited populated successor). The theorem preserves the earlier endpoint denial in its exact form: no inherited populated successor follows. What follows is stronger than silence and weaker than transported cosmography: status-existence of a fresh constraint-forced universe. Populated content, if any, is internal to a representative D_1 's own governance and is not parent-derived.

Corollary 6.5 (No-inheritance as positive force). Governance is not transmitted across B_A^* . The absence of transmission is not a defect in the theorem; it is what forces freshness. Because parent governance cannot cross and admitted nullity cannot occupy the counter-boundary register, the remaining non-degenerate status must instantiate its own kernel and AMetric licensing profile.

Corollary 6.6 (Token distinction). Any representative D_1 of the fresh internal governed-domain class is not D_0 , not a continuation of D_0 , not a duplicate of D_0 , and not a child-token containing inherited parent contents. Token distinction is forced by no-crossing and by the boundary token distinction $B_A^* \neq B_A^0$.

7 Proof Expansion and Local Proof Register

The preceding theorem gives the compressed proof. This section expands the proof into local clauses so that each possible failure point is visible. The aim is not to add new premises. It is to expose which imported theorem bears which part of the load.

7.1 The counter-boundary occupation hygiene predicate

Definition 7.1 (Counter-boundary occupation predicate). Let X be a candidate status offered relative to B_A^* . Write $\text{Occ}_{CB}(X)$ when all of the following are claimed. This is a hygiene predicate, not an existence premise:

- O1. X is not merely a parent-side sentence about the endpoint, but a counter-boundary occupation status.
- O2. X is non-null: it is not total vacuity and not mute non-classification.
- O3. X is non-degenerate: it supports admissibility-bearing evaluability, target identity, reuse, and continuation under its own scope.
- O4. X is not inherited from D_0 .
- O5. X does not require any old-scope item to cross B_A^* .
- O6. X does not use probability, metricity, causality, time, quantum state, selector, canonical representative, or operator repair as boundary authority.

Lemma 7.2 (Occupation implies kernel). If $\text{Occ}_{CB}(X)$ and X is non-degenerate, then X instantiates the AASC kernel roles (Adm , Stand , Ref , Irr) under its own domain typing.

Construction witness. Non-degenerate occupation is not raw presentation. It must distinguish licensed from unlicensed use, preserve target identity, allow reuse-stable standing, and prevent failed acts from becoming the same successful acts by later repair. These are precisely the kernel roles. If one is removed, X either loses target identity, loses standing-bearing reuse, loses the admissibility gate, or permits repair collapse. Therefore any non-degenerate occupation instantiates the kernel. $\square$

Proposition 7.3 (Kernel deletion is not a capstone survivor). Denying a kernel component does not produce a non-fresh counter-boundary status outside the route register. It destroys the register in which the capstone is stated.

Without Adm , the capstone cannot distinguish an admissible counter-boundary status from selector import, richer scope, bookkeeping, null-labeling, or post-hoc repair. Without Stand , there is no reuse-stable status whose recurrence, absence, or non-crossing can be classified. Without Ref , the boundary-pair target, the unresolved residue, and the counter-boundary candidate are not determinate. Without Irr , no-crossing and no-repair collapse because a failed transport can be relabeled as successful preservation.

The capstone therefore does not ask what follows after the kernel is abandoned. It asks what counter-boundary status remains when the kernel is preserved, old-scope routes are closed, nullity is unavailable, and no parent content crosses. Kernel deletion is not an admissible same-scope objection to the fresh governed-domain survivor; it is target destruction.

Lemma 7.4 (Inherited occupation impossible). No X satisfying $\text{Occ}_{CB}(X)$ can be inherited from D_0 .

Construction witness. Inheritance requires a parent-scope identity-preserving transport relation across B_A^* . Such a relation would carry at least one standing-bearing item, law, metric, ledger, field, record, state, or operator from D_0 as the same item. Absolute no-crossing denies all such maps. Hence inherited occupation is impossible. $\square$

Lemma 7.5 (Null occupation impossible). No X satisfying $\text{Occ}_{CB}(X)$ can be admitted nullity.

Construction witness. Admitted nullity would be a regime whose total standing-bearing content is vacuity while still carrying admissibility-bearing classification. The null-regime theorem closes that combination. If the proposed nullity is mute, it is not an occupation status and cannot satisfy O1–O3. If it is used to classify the endpoint, it is not mute and must carry non-null governance roles. Thus null occupation is impossible. $\square$

Lemma 7.6 (Boundary-authority substitutes fail). Let X be a candidate counter-boundary status whose occupation depends on probability, quantum uncertainty, primitive time, metric passage, causal bridge, a selector, or a canonical representative at B_A^* . Then X fails Occ_{CB} .

Construction witness. Each listed substitute is either downstream of governance or forbidden at the AMetric boundary. Probability presupposes a state space, alternatives, measure, and standing. Quantum uncertainty presupposes a quantum predictive interior. Time and clocks presuppose retained-order or clock-use quotients. Metricity presupposes admitted comparison. Causality presupposes witnessed dependence across an admitted order. Selectors and canonical representatives violate the no-selector/no-parameter AMetric theorem. None can authorize counter-boundary occupation before governance. $\square$

7.2 Clause-by-clause proof register

Proof clause	Load-bearing import	Local conclusion
Boundary exposure	New AMetric Boundary theorem	R_H closes as B_A^* , not singularity, storage, transfer, bounce, or nullity.
Profile homology	AMetric Normal-Form Homology	$\text{PANF}(B_A^*) = \text{PANF}(B_A^0) = \mathbf{aANF}$.
No transmission	Absolute no-crossing	No parent-scope content can serve as counter-boundary governance.
Null closure	Null Regime theorem	Admitted vacuity is not a terminal regime.
No selector	AMetric boundary theorem	Boundary cannot choose, rank, index, probabilize, or metricize the far side.
No repair	Operator closure and kernel irreversibility	Old-scope partiality cannot be totalized into successor transport.
No primitive time	Arrow and Clock theorems	Before/after and clock-selection language cannot authorize the boundary.
No metric bridge	Metricity licensing	Metric continuation cannot cross an AMetric boundary.

No rival interiors	Unus Solus	Same-scope multiple interiors collapse; token distinction requires fresh internal governed-domain status.
Survivor identification	Kernel necessity plus route exhaustion	Fresh internal governed-domain status is the unique non-degenerate status-class up to $\equiv_{\text{gov},CB}$.

7.3 Why this is not causal production

A causal-production theorem would supply a mechanism G such that $G(D_0) = D_1$, or a dynamical relation carrying data from D_0 into D_1 . This theorem supplies neither. It proves something cleaner: once every non-fresh alternative is closed, the only remaining non-degenerate counter-boundary status-class is a fresh internal governed domain. The absence of a production mechanism is not a weakness, because production would require the temporal, causal, metric, and transport structures that the AMetric boundary excludes.

7.4 Why no-inheritance strengthens the universe claim

No-inheritance is the theorem's positive condition for fresh internal governed-domain status. It gives the claim its exact AASC form. Since no parent law, metric, information, energy, field, particle identity, record, or clock crosses B_A^* , the new domain cannot be an inherited successor. Since nullity, ungoverned residue, boundary-only terminality, probability, primitive time, metric passage, causal production, selector routes, same-scope repair, and multiple rival interiors are closed, the counter-boundary register cannot remain empty or arbitrary. The only surviving status is fresh constraint-governed interiority: a universe by kernel minimality, not by parent transmission.

7.5 Why this is stronger than agnosticism

Agnosticism would say that, given no-crossing, nothing further can be classified. That is incomplete because nullity, ungoverned residue, inherited continuation, selector routes, and same-scope repair are all not merely unknown; they are closed. The theorem is therefore neither detailed cosmography nor silence. It is the forced normal form for counter-boundary status: a fresh internal governed domain without inheritance.

8 Detailed Closure of Boundary-Authority Routes

This section records the most common false routes in more detail. The target is to prevent a later reader from smuggling an interior structure into the boundary under a familiar name.

8.1 Probability and typicality

Probability is often used as a soft selector: if many outcomes are possible, the boundary might sample one. Under AASC this is unavailable at B_A^* . A probability measure requires an admitted outcome space, identity of alternatives, measurable structure, comparison across alternatives, and standing

for the measure itself. All of these are governance-bearing structures. At an AMetric boundary they cannot function as boundary authority. A probability distribution can be declared inside a richer third-domain model, but then that model supplies independent governance rather than parent-scope boundary selection.

Proposition 8.1 (No probability-first counter-boundary selection). There is no admissible same-scope route from B_A^* to D_1 by probability-first selection.

Construction witness. If the probability is parent-scope, it would have to cross. If it is boundary-supplied, it violates AMetric non-parameterization. If it is third-domain, it changes scope. If it changes no standing-bearing status, it is bookkeeping. Thus no probability-first selection remains. $\square$

8.2 Quantum uncertainty and tunneling

Quantum uncertainty is not pre-governance indeterminacy. It is a physical interior structure involving states, amplitudes, operators, records, and probability rules. Tunneling is likewise a model-internal amplitude statement. It presupposes a Hamiltonian or path-integral structure, boundary conditions, and a state space. These are not available as AMetric boundary authority.

Proposition 8.2 (No quantum-first crossing). A quantum description can avoid the theorem only by denying an antecedent or by declaring richer governance before endpoint closure. It cannot serve as a same-scope crossing route through B_A^* merely by invoking uncertainty, tunneling, or global unitarity.

8.3 Temporal sequence and causal production

Questions such as *what happens after the boundary?* and *what caused the new regime?* are external-use language unless a governed comparison domain is declared. Time is an interior retained-consequence order. Causality is witnessed dependence across admitted order. Neither has standing at the AMetric boundary. The theorem therefore uses *counter-boundary* as a classification relative to boundary token, not as temporal succession.

Proposition 8.3 (No primitive before/after relation). There is no parent-scope temporal order placing D_1 after D_0 across B_A^* . Any such ordering is either an external description without standing-bearing force, a third-domain relation, or a scope change.

8.4 Metric and causal bridge

A metric bridge would supply distance, duration, continuity, curvature, or topology across B_A^* . A causal bridge would supply dependence across the boundary. Both require exactly the interior roles that the boundary excludes. If a bridge is declared by a richer theory, that theory must be classified as a third-domain or new governance scope; it is not inherited parent-scope continuation.

8.5 Selector, canonical representative, and branch label

Selectors and representatives are especially tempting because they can look harmless. But if a branch label or representative changes counter-boundary status, it is boundary-authoritative and prohibited. If it changes nothing, it is skin. If it is supplied by a larger theory, it belongs to richer scope. Thus no branch label can do the work of fresh internal governed domainhood.

9 Structural Identity Without Inheritance

No-crossing explains the exact form of the universe claim. Since nothing parent-scope can cross, the recurrence cannot be genealogical, transmissive, or causal. It must be structural. The counter-boundary regime is role-equivalent to any origin-licensed governed-interior architecture because the AASC kernel and AMetric normal form are the minimal conditions for non-degenerate governed domainhood. This section sharpens that distinction.

9.1 Role recurrence versus token inheritance

Definition 9.1 (Role recurrence). Role recurrence occurs when two token-distinct regimes instantiate the same required role because both satisfy the same target-class necessity. It is not transmission.

Definition 9.2 (Token inheritance). Token inheritance occurs when the same standing-bearing item, law, metric, record, field, or identity package persists from one regime into another. It requires crossing and is prohibited here.

The counter-boundary regime may instantiate admissibility, standing, reference, irreversibility, locality, metric licensing, field carriers, gauge structure, or Standard-Model structural roles if the corresponding target class is fixed. None of those token identities is inherited from D_0 . They recur because the target role is forced, not because the parent token survives.

Lemma 9.3 (Role identity is not law-token isomorphism). Role identity at the kernel and AMetric level forces the minimal governance architecture. It does not by itself assert sameness of concrete law tokens, numerical constants, fields, particle inventory, histories, records, metric realization, or empirical population. Any stronger isomorphism of physical law packages requires realized-physical-interior and realized-standard-physics target fixation plus whatever exactness certificates govern those layers. The present theorem forces the universe-licensing architecture, not inherited cosmography.

Construction witness. Kernel and AMetric identity fix admissibility, standing, reference, irreversibility, no-repair, no-selector, no-primitive-metricity, and independent target gating. These are governance coordinates. Law tokens, parameter tables, histories, fields, and populations are downstream occupants or structures internal to a governed domain. Since no-crossing blocks parent-token inheritance, role equivalence cannot license token identity. $\square$

9.2 Parameter discipline

The same distinction controls constants. A structural slot can be forced without fixing the same numerical coordinate. A gauge role can be forced without inheriting a gauge-coupling value. A mass-response slot can be forced without inheriting a mass table. A clock role can be forced without inheriting a parent clock. A vacuum baseline can be forced without inheriting a cosmological constant value.

Object	Role-level status	Parameter/content status
Admissibility gate	Forced in any non-degenerate regime.	Parent gate token not inherited.

Metric role	Forced if realized physical interior target is fixed.	Parent metric not inherited.
Gauge structure	Forced under realized-standard-physics target.	Coupling values not inherited.
Mass-response role	Forced under physical interior and SM specialization.	Exact masses not inherited.
Vacuum baseline	Forced as non-null no-particle baseline.	Vacuum-energy numerical ledger not inherited.
Time/order role	Forced as retained-consequence ordering in physical interior.	Parent time not inherited.
Minimal life-carrier form	Further conditional under fixed-domain life antecedents.	No inherited life, molecular substrate, intelligence, technology, or abundance claim.

9.3 Reporting standard

A report about D_1 must say which layer it occupies:

1. kernel necessity;
2. AMetric boundary discipline;
3. realized-physical-interior role necessity;
4. standard-physics structural specialization;
5. further conditional life-carrier target-class specialization;
6. numerical parameter derivation;
7. concrete populated content.

Only the first two are unconditional for non-degenerate counter-boundary governance. The third and fourth are forced once their target-class antecedents are fixed. The fifth is a further conditional downstream specialization requiring fixed life-domain antecedents. The sixth and seventh are not inherited and are not supplied by this theorem.

10 What Is Forced Without Crossing

The no-crossing theorem blocks inherited content. It does not block necessity. AASC distinguishes *role-level force* from *parent-derived populated content*. The former can be known without crossing because it follows from kernel minimality, AMetric boundary discipline, and target-class antecedents; the latter cannot be inherited.

Layer	Status	Knowability rule
Kernel	Unconditional for non-degenerate occupation	Admissibility, standing, reference, irreversibility, bivalence, no repair, and standing conservation are forced.

AMetric boundary	Unconditional	No primitive metric, time, probability, selector, representative, inherited law, or operator has boundary authority.
Realized physical interior	Conditional on target fixation	Locality, metric licensing, field carriers, particle persistence classes, gauge structure, mass/inertia, vacuum, time, causality, and records are role-forced if D_1 is fixed as realized physical.
Standard-physics structural layer	Conditional on realized-standard-physics target	Gauge product, representation roles, Higgs/bridge, Yukawa slots, charge lattice, anomaly neutrality, vacuum compatibility, and parameter-status discipline are role-forced.
Minimal life-carrier target	Further conditional specialization	Minimal carrier structure is forced only when fixed-domain life antecedents are satisfied: bounded carrier family, recurrent windows, local repair, retained identity-memory, standing margin, standing localization, and finite standing-fixation interval.
Intelligence / technology	Not part of the main theorem	Only conditional standing-margin or interface-budget constraints if separately targeted; no intelligence inevitability, technological inevitability, SETI, or Fermi conclusion is derived here.
Parameter/content layer	Not supplied by this theorem	Numerical constants, exact masses, histories, records, occupants, and empirical contents are non-inherited but internally constrained by anchors, witnesses, calibration, quotient stability, and no-repair discipline.

10.1 Unconditionally forced by kernel necessity

Any non-degenerate counter-boundary regime must instantiate:

- admissibility as fail-closed gate;
- standing as reuse-stable admissibility;
- reference as target fixation;
- irreversibility as no same-act repair;
- bivalent standing on the evaluated fragment;
- conservation of standing: preservation or destruction, not internal creation, transfer, amplification, or repair;
- uniqueness of the admissible interior up to lawful equivalence;
- no primitive selector, probability, metric, clock, representative, or inherited law at the boundary.

10.2 Unconditionally forced by AMetric boundary discipline

At the counter-boundary AMetric side, the following are inadmissible as boundary authority:

- metric distance, duration, curvature, scale, and coordinates;
- causal order and temporal before/after relation;
- probability measure, typicality rule, and quantum uncertainty;
- canonical representative, branch label, index, or selector;
- inherited law, inherited constant, inherited field, inherited state, or inherited record;
- operator domain, repair mechanism, transfer map, or totalization procedure.

10.3 Conditionally forced by realized-physical-interior necessity

If D_1 is fixed as a realized physical interior, then the role architecture is richer than the bare kernel. The realized physical interior theorem forces the following roles at the role level:

Role	Forced function
Non-vacuous realization	The regime must support non-null standing-bearing persistence rather than mute presentation.
Locality	Standing-bearing content must localize sufficiently to carry comparison, perturbation, transport, and retained continuation.
Metric licensing	Metricity may appear only as licensed interior comparison, not boundary authority.
Field carriers	Persistent mediation roles must carry standing across local continuation and interaction.
Particle persistence classes	Localized persistence classes must stabilize identity under interaction and redescription.
Gauge structure	Boundary-fixed presentation equivalence must organize local representatives without promoting skin to tensor.
Mass/inertia response	Deformation-response and inertial roles must mediate persistence under perturbation.
Vacuum baseline	A non-null no-particle baseline is required for field and particle roles.
Time	Time is irreversible continuation ordering inside the interior, not primitive boundary flow.
Causality	Causality is witnessed dependence across admitted order, not primitive influence.
Records	Records are fixed witnesses available for downstream reuse.

These are not inherited from D_0 . They are forced by the target *realized physical interior*. If D_1 is not fixed as a realized physical interior, these roles are not locally asserted. If it is so fixed, they are known by necessity without crossing.

10.4 Conditionally forced by Standard-Model structural recovery

If the target class is further fixed as the same realized-standard-physics structural class, then the Standard Model structural envelope is forced at the role level. This includes:

- gauge-product structure;
- representation-response roles;
- Higgs/bridge fixation role;
- Yukawa-realizability slots;
- charge-lattice rigidity;
- anomaly neutrality;
- vacuum compatibility;
- mass-response and mixing slots;
- parameter-status discipline separating structural closure from numerical parameter derivation.

This is not inheritance of the parent Standard Model token. It is target-class structural recovery. The numerical values of gauge couplings, masses, mixing angles, Higgs-sector coordinates, and other empirical constants remain downstream parameter targets unless a separate exactness certificate, calibration ledger, or reconstruction rule is supplied.

10.5 Further conditional specialization: life-carrier target class

The life-carrier result is used only at the conditional status declared in the target-class hierarchy lock. If a fresh domain is already fixed as a realized physical interior and also satisfies the fixed life-domain antecedents, then minimal life-carrier structure recurs by the same admissibility logic.

The relevant antecedents are a fixed physical domain, an admissible resolution, a bounded-carrier family, recurrent windows, non-degenerate perturbation burden, local repair, retained identity-memory, standing localization, and a finite standing-fixation interval. Under those antecedents, minimal carrier form is forced. Without those antecedents, the black-hole theorem does not report life-carrier status.

10.6 Not inherited from the parent side

The following are not parent-derived:

- numerical constants and exact coupling values;
- exact masses and mixing coordinates;
- metric dimension, topology, or global geometry;
- concrete law tokens;
- particle inventory as populated fact;
- histories, records, observers, memories, or empirical contents;
- parent information, parent energy, parent field identity, parent quantum state, or parent metric;
- any causal relation from D_0 to D_1 .

The next section strengthens this classification. The forced-without-crossing ledger does not merely identify which layers are non-inherited. It also identifies which layers recur by constraint symmetry and which downstream contents remain variable only under internal certification.

11 Minimal Constraint Identity from Homologous Licensing

The homology theorem already establishes role-profile equality:

$$\text{PANF}(B_A^*) = \text{PANF}(B_A^0) = \text{aANF}.$$

Homology is not transmission. But once no-crossing is fixed, transmission is not the relevant route. Homology matters because it identifies the endpoint boundary as the same licensing condition in role-profile form. Kernel minimality then supplies the force: any non-degenerate occupation of that condition instantiates the same governing roles. The result is not inherited sameness; it is constraint identity.

Theorem 11.1 (Constraint identity from homologous licensing). If $\text{PANF}(B_A^*) = \text{PANF}(B_A^0) = \text{aANF}$, no-crossing holds, admitted nullity is closed, and non-governed residue is inadmissible, then any non-degenerate counter-boundary domain is role-equivalent to any origin-licensed governed-interior architecture at the level of minimal governance roles. This identity is not inherited; it is forced by the minimality of the kernel.

Construction witness. The endpoint boundary has the same complete AMetric role profile as the origin boundary. The boundary supplies no selector, probability measure, inherited law, metric, time, causal order, operator domain, or transport relation. A non-degenerate domain cannot be ungoverned, because non-degenerate domainhood requires admissibility, standing, reference, and irreversibility. It cannot be null, because admitted nullity is closed. Therefore any non-degenerate counter-boundary domain must instantiate the same kernel and AMetric role conditions as any origin-licensed governed-interior architecture. Since no parent-scope item crosses, the identity cannot be inheritance. It is role equivalence by minimal constraint. $\square$

Corollary 11.2 (Origin and endpoint are bookends by form only). Origin and endpoint are structurally homologous by boundary normal form. They are not the same token, not temporally linked, not cyclically identified, and not physically connected by a crossing map. The universe-level recurrence is therefore not recurrence of an old universe, but recurrence of the licensing condition that forces a fresh governed one.

12 Constraint Symmetry and Downstream Non-Arbitrariness

The Counter-Boundary Universe Theorem does more than establish that a fresh internal governed domain is forced. It also fixes the kind of sameness such a domain has, and the kind of variation it can still admit. The sameness is not genealogical. It is not law inheritance, metric inheritance, field inheritance, parameter inheritance, history inheritance, biological inheritance, or token reproduction. It is constraint identity: independent instantiation of the same minimal role architecture required for non-degenerate governed domainhood.

12.1 Status-class result versus token instantiation

The main theorem is a status-class theorem. It proves the unique non-degenerate counter-boundary status-class under the certified antecedents. It does not select a populated token domain, specify a concrete history, enumerate occupants, or assert a plurality of token cosmologies. Constraint-symmetry corollaries apply to representative occupants only under additional target-class fixation.

This section records the implication directly. A fresh universe is not copied from the parent, but if it is instantiated as a realized physical interior, it is forced into the same role architecture as any non-degenerate realized physical interior. If further target classes are audited and fixed—for example the minimal Lorentzian metric projection, the realized-standard-physics structural envelope, or a separately declared fixed-domain life-carrier target—their structural forms recur conditionally. Numerical constants, histories, empirical records, law tokens, and populated contents remain non-inherited, but they are not free variables. Any such content must be internally certified by the fresh domain’s own admissibility and standing discipline.

Target-class fixation is not a switch that makes any recurrence claim true. Each target must declare its antecedents, carriers, failure modes, comparison class, and same-scope alternatives before it can support a role-recurrence corollary.

Definition 12.1 (Constraint identity). Two token-distinct domains are constraint-identical at level L when they independently instantiate the same minimal role architecture required by the same non-degeneracy conditions, without crossing, inheritance, token identity, causal production, common contents, or shared parent-scope history.

Definition 12.2 (Role-exact recurrence). A fresh domain D exhibits role-exact recurrence relative to a target class T when, if D satisfies the antecedents of T , it instantiates the same forced role package as any other non-degenerate occupant of T , up to admissible redescription.

Definition 12.3 (Downstream content). Downstream content means numerical constants, parameter values, empirical records, histories, concrete field occupants, law tokens, scale displays, calibration values, populated contents, and other internal items that appear only after the governance and target-class roles have standing.

Definition 12.4 (Internal certification). A downstream content item has internal certification in a fresh domain D only when it is licensed by D ’s own admissibility, standing, reference, quotient, witness, calibration, and no-repair conditions. Internal certification is not inheritance; it is the domain-local standing of the item.

Theorem 12.5 (Realized-interior constraint symmetry under target fixation). For any token-distinct representative occupants of the fresh internal governed-domain status-class, if the same realized-physical-interior target-class antecedents are fixed, the forced realized-physical role package is equivalent up to admissible redescription. If the same realized-standard-physics target-class antecedents are also fixed, their standard-physics structural envelopes are equivalent up to admissible redescription. Differences in numerical constants, histories, records, field occupants, concrete law tokens, or parameter displays do not break this equivalence unless they alter a standing-bearing role ledger.

Construction witness. Fresh internal governed domainhood already forces the kernel and AMetric licensing conditions. The realized-physical-interior target adds the minimal role package required for standing-bearing physical persistence. The realized-interior theorem states that this package is coherent, closed, and operationally minimal: deletion of a major role destroys the target or changes scope [12]. Therefore two fresh domains satisfying that same target cannot differ at the forced-role level without losing the target. If the realized-standard-physics target is additionally fixed, the Standard Model structural theorem supplies the recovered structural tuple as the admissible survivor of the gauge, representation, bridge, charge, anomaly, vacuum, mass, and mixing role constraints [13]. Differences below those role ledgers may remain, but only if they are internally certified and do not change the standing-bearing structure being compared. Thus the symmetry is constraint-level symmetry, not token identity or inheritance. $\square$

Proposition 12.6 (Realized-interior recurrence under target fixation). If a fresh counter-boundary domain D_1 is fixed as a realized physical interior, then it must instantiate the realized-physical-interior role tuple up to admissible equivalence. This tuple consists of non-null realization; bounded standing-localization; adjacency, neighborhood, and local-continuation roles; quotient-stable comparison; bounded perturbation transport; field-carrier roles; particle-persistence classes; gauge roles; mass/inertia roles; vacuum baseline; temporal ordering; and causal-dependence roles.

Construction witness. The realized physical interior theorem forces non-null realization, bounded standing-localization, local continuation domains, quotient-stable comparison, bounded perturbation transport, fields as persistence carriers, particles as localized persistence classes, gauge as boundary-fixed presentation equivalence, mass/inertia as deformation-response roles, vacuum as a non-null no-particle baseline, time as irreversible continuation ordering, and causality as witness-dependent dependence [12]. A regime lacking one of these roles is not a radically different realized physical universe in the same target class. It changes target, imports hidden load, collapses to skin, or fails non-degeneracy. $\square$

Proposition 12.7 (Standard-physics structural corridor under target fixation). If a fresh realized interior is further fixed as a realized-standard-physics target, then the same structural corridor applies: gauge-product structure, representation-response skeleton, Higgs/bridge carrier, Yukawa-realizability ledger, charge lattice, anomaly-neutral survivor, vacuum-compatible residual form, mass/mixing slots, and family-cardinality envelope are forced up to admissible redescription. This is a target-class recurrence theorem, not a universal theorem that every fresh internal governed domain instantiates the Standard Model structural envelope. It is not used to prove fresh universe-status; it classifies the structural recurrence that follows only after the realized-standard-physics target class is separately fixed.

Construction witness. The standalone Standard Model structure paper separates structural derivation from the downstream empirical parameter table. Its recovered tuple records the gauge product, representation-response roles, Higgs-like bridge/fixation carrier, Yukawa-realizability and coupling-slot ledger, charge lattice, anomaly-neutral survivor ledger, vacuum-compatible residual form, mass-response slots, mixing/phase slots, and family-cardinality envelope record as structural entries [13]. Therefore a fresh domain satisfying the same target class is not free to vary those structural ledgers without changing the target or introducing richer-scope load. Numerical values inside the slots remain downstream. $\square$

Corollary 12.8 (Einstein-form recurrence under metric-target fixation). If a fresh domain instantiates the AASC gravity role and is faithfully projected into the same minimal Lorentzian first-response metric target, then the Einstein-form projection recurs:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

The recurrence is form-level and role-level. It is not recurrence of numerical κ , numerical Λ , matter-sector scale, or all gravitational dynamics. Numerical values require internal scale, source, and calibration ledgers.

Construction witness. The Einstein projection theorem proves the displayed equation as the unique minimal first-response same-scope metric projection of AASC gravity under a faithful Lorentzian metric target and closed source continuation [15]. The theorem also treats dimensional constants by scale-import and calibration discipline rather than boundary generation. Hence a fresh domain with the same metric target recurs in Einstein form without inheriting the parent's metric token, source ledger, cosmological history, or numerical constants. $\square$

Proposition 12.9 (Downstream non-arbitrariness). Let x be a numerical constant, empirical record, history, parameter value, law token, calibration value, or concrete occupant inside a fresh domain D_1 . If x has standing in D_1 , then x must be internally licensed by D_1 's own admissibility structure. It must occupy a standing-bearing role or slot, survive lawful redescription, respect quotient identity, satisfy the relevant witness or calibration ledger, avoid hidden selector import, and preserve no-repair discipline.

Construction witness. A downstream content item with standing must be evaluable, reference-fixed, reusable under lawful redescription, and protected from same-act repair. Otherwise it is skin, hidden tensor load, representative privilege, post-hoc repair, or scope change. Non-inheritance prevents the parent from supplying x . It does not exempt x from admissibility inside D_1 . Parameter-level non-inheritance therefore means internal certification is required, not that parameter-level content is free. $\square$

Corollary 12.10 (Minimal life-carrier recurrence under fixed-domain life antecedents). Let D_1 be a representative occupant of the fresh internal governed-domain status-class already fixed as a realized physical interior. If D_1 also satisfies the fixed life-domain antecedents—a fixed physical domain Ω , admissible resolution $\varepsilon_0 > 0$, bounded carrier family B_{ε_0} , recurrent windows, non-degenerate perturbation burden, local repair conditions, retained identity-memory, standing localization, and finite standing-fixation interval $T_{\text{fix}} < \infty$ —then D_1 instantiates minimal life-carrier structure in the fixed-domain sense. This recurrence is not biological inheritance, not abiogenesis prediction, not population transfer, and not a claim about molecular substrate, intelligence, technology, detectability, or empirical abundance.

Construction witness. The fixed-domain life-carrier theorem proves forced locality of repair, retained repair $R_e = JR$, collapse of admissibility-relevant viability to the standing margin $\Sigma = R_e - D$, global exclusion of non-local standing, strict minimality of minimal carriers under resolved inclusion, and minimal-carrier inevitability within the fixed life-domain theorem when the fixed antecedents and finite T_{fix} obtain [14]. Those are additional life-domain antecedents beyond the black-hole theorem. If they are not fixed, no life-carrier status is reported by the present manuscript. If they are fixed, minimal carrier recurrence follows by the same target-class constraint logic that governs realized-interior and standard-physics recurrence. $\square$

Life-domain antecedent	Function
Fixed physical domain Ω	Prevents cross-domain biological projection.
Admissible resolution ε_0	Supplies the resolved bounded-carrier family.
Bounded carriers B_{ε_0}	Fixes the carrier comparison class.
Recurrent window $W = (t, \tau)$	Supplies the persistence interval.
Perturbation burden $D_B(W)$	Supplies degradation/load condition.
Identity-memory $J_B(W)$	Supplies retained identity condition.
Local repair $R_B(W)$	Supplies same-carrier repair condition.
Standing margin $\Sigma_B(W) = R_e - D$	Supplies viability and standing gate.
Finite T_{fix}	Supplies minimal-carrier existence condition.

No black-hole-to-life inference. The black-hole theorem alone does not imply life. It implies fresh internal governed-domain status. The realized-interior recurrence theorem then constrains physical role architecture when that target is fixed. The life-carrier theorem applies only after a further fixed-domain life target is declared. Thus the manuscript does not infer life from black holes; it classifies what life-carrier standing would require if the fresh domain satisfies the relevant life-domain antecedents.

Forced role level	Recurrence status
Kernel	Role-exact under non-degeneracy.
AMetric normal form	Role-exact at licensing profile.
Fresh internal governed-domain status	Role-exact status-class survivor under governance equivalence.
Realized physical interior	Role-exact under realized-physical target fixation.
Minimal Lorentzian metric projection	Role-exact form under metric-target fixation.
Standard-physics structural envelope	Role-exact structural corridor under target fixation.

Downstream layer	Status
Numerical constants, records, histories	Non-inherited but internally certified.
Minimal life-carrier target	Further conditional on fixed-domain life antecedents.
Intelligence	Not forced; conditional standing-margin augmentation if separately targeted.
Technology	Not forced; conditional interface-budget persistence if separately targeted.
Populated contents	Not supplied by this theorem.

Remark 12.11 (Meaning of structural near-identity). The controlled statement is not that every fresh domain has the parent’s parameter table, empirical history, or concrete occupants. The controlled statement is stronger at the correct level: role architecture recurs role-exactly under matched target conditions, while downstream contents vary only where the new domain internally certifies that variation. In theorem language, fresh universes are role-exact at the forced-architecture level; allowed difference is pushed below that level and remains subject to internal non-degeneracy constraints.

13 Black-Hole Application

The theorem applies to certified AASC black-hole targets only under the antecedents. It is not an empirical attainment theorem for every astrophysical compact object.

13.1 Decision table

Case	Condition	Result
No unresolved burden	$R_H = \emptyset$.	Endpoint theorem does not activate; exterior witness resolution has discharged the burden.
Partial residue, no exhaustion	$R_H \neq \emptyset$ but residual support has not reached maximum governance-depth.	Residual-coherence analysis applies; New AMetric Boundary not yet exposed.
Certified re-governance before exhaustion	A richer independently governed scope is declared before residual metric support exhausts.	The richer scope must be classified on its own terms; no old-scope crossing follows.
Maximum governance-depth	$R_H \neq \emptyset$, residual metric support exhausts, nullity and false endpoints closed.	B_A^* is exposed and the Counter-Boundary Universe Theorem applies.
Uncertified permanent horizon	No leakage, boundary trace, holographic anchor map, remnant map, non-sink quotient, or witness relation resolves the burden.	Exterior reconstruction exhaustion and unresolved residue formation.

13.2 Information paradox retyping

The theorem does not say that information is destroyed. It says the parent-scope question *where did the information go?* is ill-typed after no-crossing if *information* means the same parent-scope standing-bearing carrier package. A Page curve, entropy ledger, or radiation flux resolves only the burden it actually re-fixes through a witness relation. Otherwise the unresolved burden remains in R_H until endpoint closure.

13.3 Singularity retyping

A singularity is a metric-diagnostic statement internal to a still-operative metric articulation. At maximum governance-depth, the residual role-support under which metric articulation can diagnose a limit has ended. The endpoint is therefore not a more extreme singularity. It is a New AMetric Boundary.

13.4 Energy and first-law retyping

Parent-domain energy and first-law ledgers remain valid within the parent scope. They do not cross. No-crossing is not nonconservation; nonconservation is a ledger-bearing interior statement. The parent ledger cannot be balanced by borrowing a counter-boundary domain without declaring a third-domain comparison scope.

14 Program Interfaces

14.1 Quantum gravity

A quantum-gravity proposal can respond in four admissible ways:

1. deny target certification;
2. resolve R_H through a genuine exterior witness relation before maximum governance-depth;
3. prevent maximum governance-depth by supplying independently declared richer governance;
4. accept B_A^* and the no-crossing/fresh-internal-governed-domain classification.

A proposal that merely names unitarity, entropy flow, a global state, a boundary slogan, or a tunneling amplitude without target-fixing standing transport does not meet the counterexample burden.

14.2 Holography and islands

Holographic and island descriptions matter only through the maps they supply. A slogan is skin. A calculational device is not automatically a witness relation. A genuine map must re-fix the threatened reference-standing classes under the declared exterior quotient. If it does, it subtracts from R_H . If it does not, R_H remains.

14.3 Remnants, fuzzballs, final states, and baby-universe comparators

A remnant without an exterior standing map is storage language without a witness. A fuzzball model is a richer tensor package only if its microstate structure is declared and linked to the standing burden. A final-state projection introduces boundary conditions and must be typed as declared governance, not same-scope repair. Baby-universe language is an external comparator unless it supplies a governed target. The present theorem permits the ordinary-language slogan only under AASC discipline: certified endpoints force fresh internal governed domains by constraint recurrence, not by baby-universe tunneling, inherited law, causal production, or parent-content transfer.

14.4 Cosmology

The theorem is a universe-recurrence theorem, not a cyclic-cosmology theorem. A certified AASC black-hole endpoint does not send this universe into another, recycle parent matter, preserve parent law, transfer entropy, or transport information. It exposes a homologous AMetric licensing condition. Since the kernel is regime-invariant, that licensing condition forces a fresh governed interior. Thus the formal result is universe-recurrence only in the AASC-admissible sense: a new constraint regime is forced; an old one is not transferred. A fresh regime does not inherit a Past Hypothesis, inflationary mechanism, low-entropy condition, constants, laws, or cosmic history from D_0 .

14.5 Constants and parameter discipline

The theorem permits role-level structural recurrence without numerical inheritance. If a realized-standard-physics target is fixed, certain structural slots recur by necessity. Exact parameter values do

not cross. A coupling value, mass, mixing angle, vacuum scalar, cosmological parameter, or dimensional scale has standing only inside an admitted regime with its own calibration and witness package.

14.6 Realized-interior and standard-physics structural recurrence

A fresh-domain recurrence claim can be misunderstood in two opposite ways. One error says that nothing crosses and therefore the fresh domain is arbitrary. The other says that structural recurrence must be hidden inheritance. AASC blocks both readings. The Realized Physical Interior theorem supplies the conditional role package for realized-interior recurrence: non-null realization, localization, local domains, comparison, bounded transport, field carriers, particle persistence classes, gauge structure, mass/inertia response, vacuum baseline, time, causality, and records as standing-bearing roles. The Einstein projection theorem supplies the conditional metric-form recurrence under faithful minimal Lorentzian first-response target fixation. The Standalone Standard Model theorem supplies the conditional standard-physics structural corridor and parameter-status discipline. The Minimal Life Carriers theorem supplies a further conditional downstream specialization only when fixed-domain life antecedents are declared. None of these imports parent contents into the fresh domain. They state what must recur when the corresponding target class is fixed.

15 Interface with Standard Black-Hole Literature

The theorem is positioned against standard black-hole programs as a classification theorem, not as a replacement calculation. Bekenstein-Hawking entropy and Hawking radiation remain parent-domain thermodynamic and semiclassical ledgers. Page-curve behavior is relevant when paired with a target-fixing reconstruction map, but an entropy profile by itself is not a standing witness. Holographic duality, quantum extremal surfaces, replica wormholes, and island formulas are treated as possible certificate structures only where their maps re-fix the threatened reference-standing classes under the declared exterior quotient. Remnants, fuzzballs, final-state projections, and baby-universe sectors are likewise not rejected by name; they are typed as witness maps, declared tensor load, richer governance, or insufficient storage language depending on the role they actually perform.

The interface burden is therefore explicit: a standard proposal challenges the theorem only by denying the certified target, resolving R_H before endpoint closure, preventing maximum governance-depth by declared richer governance, or supplying a same-scope admissible status outside the route register. Familiarity, calculational power, entropy behavior, or formal elegance is not by itself a standing-preserving witness relation.

16 Objection Matrix

Objection	Reply
This uses a provocative ordinary-language slogan.	The formal AASC reading is fresh internal governed-domain status under certified endpoint antecedents. The transmissive baby-universe reading is rejected: no parent contents, laws, metrics, information, or energy cross.

This contradicts the no-populated-successor line.	No. That line denies inherited populated successorhood. The present theorem preserves it and adds the stronger result that a fresh internal governed domain is forced once non-fresh alternatives close.
Nothing crosses, so nothing follows.	No. Nothing crosses; therefore inheritance fails. Once nullity and non-fresh alternatives close, fresh constraint-governed recurrence follows.
The theorem is merely definitional.	No. The route register is closed by imported fixed-domain exhaustion, null-regime closure, no-selector boundary discipline, no-repair, and uniqueness. A counterexample must supply a faithful status-class survivor.
Nullity remains possible.	Admitted nullity is closed. Mute nullity is not an admissible regime and carries no classificatory force.
Quantum probability can select the outcome.	Probability requires state space, measure, alternatives, and standing. It is licensed interior structure, not boundary authority.
Quantum tunneling crosses forbidden boundaries.	Tunneling is defined inside a quantum model with amplitudes, states, and boundary conditions. It is not pre-governance crossing through an AMetric boundary.
A global wavefunction contains both sides.	A global wavefunction is a richer-scope object. It must declare its own governance and cannot serve as old-scope crossing without a target-fixing transport map.
A third-domain theory relates both domains.	It may, but then it is third-domain governance, not same-scope crossing. It cannot refute no-crossing unless it preserves the theorem target.
Energy must be conserved across the boundary.	Conservation is a ledger-bearing statement. Cross-AMetric conservation requires a shared ledger, clock, comparison carrier, and source identity, all of which are unavailable at the boundary.
If energy does not cross, it is destroyed.	Destruction is an event classification inside a temporal-causal ledger. The theorem says the predicate lacks cross-boundary conditions.
If energy appears in D_1 , it was created.	Creation is likewise a temporal-causal event classification. D_1 's source roles, if any, stand under its own governance.
The endpoint causes D_1 .	Causality is an interior role. The theorem supplies constraint-forced occupation by elimination, not causal production.
The boundary must be at the singularity.	A singularity is a metric diagnostic. The New AMetric Boundary is where metric-diagnostic authority has exhausted for R_H .
A larger QG theory could avoid this.	Yes, by denying an antecedent: resolving R_H , preventing exhaustion, or declaring richer governance before endpoint closure.

This is unfalsifiable.	The theorem is target-class conditional, and target-class fixation is audited rather than stipulated. It can be challenged by denying target certification, breaking an import, showing dependency misuse, proving antecedent incompatibility, or exhibiting a faithful status-class survivor.
Standard Model recurrence exceeds the theorem.	It claims role-level structural recovery only under a fixed realized-standard-physics target class. It does not inherit numerical constants or parent empirical history.
This collapses no-inheritance into sameness.	No. No-inheritance blocks parent transmission. Constraint symmetry says that independently instantiated non-degenerate domains solve the same minimal role problem. The sameness is necessity-level, not genealogical.
If parameters can differ, the interiors are not structurally identical.	Parameter variation does not defeat role equivalence unless it changes a standing-bearing role ledger. Raw numerical displays are skin unless internally certified; structural identity is evaluated at the kernel, realized-interior, metric-projection, and structural-envelope levels.
The paper claims every fresh universe has the Standard Model.	No. The claim is conditional. If the fresh domain is fixed as a realized-standard-physics target, the Standard Model structural corridor recurs. The black-hole theorem alone proves fresh internal governed-domain status.
Parameters are arbitrary because nothing crosses.	No. Nothing crosses from the parent. But any parameter or record inside the fresh domain must be internally licensed by that domain's own admissibility and standing conditions.
Life recurrence exceeds the theorem.	The life-carrier claim is a further conditional target-class result requiring fixed-domain life antecedents: bounded carriers, recurrent windows, local repair, retained identity-memory, standing margin, localization, and finite standing-fixation interval. It is not biological inheritance.
The theorem infers life from black-hole endpoints.	No. The black-hole theorem proves fresh internal governed-domain status. Life-carrier recurrence is a further conditional target-class result requiring fixed-domain life antecedents. No biological inheritance, abiogenesis probability, intelligence, technology, detectability, or empirical abundance is inferred from the black-hole theorem.
The life layer is speculative biology.	The life-carrier import is not biochemical or probabilistic. It is a fixed-domain standing-localization theorem: local repair, retained identity-memory, standing margin, localization, and strict minimal carrier descent. Molecular substrate and empirical abundance are outside the claim.

17 Counterexample Burden

A counterexample must preserve the target. It must not merely ask for a different theorem class.

Black-hole counterexample burden after quotient subtraction. A claimed black-hole counterexample must preserve the endpoint handoff while avoiding the capstone classification. It must therefore exhibit a nonempty unresolved residue after all exterior quotient and certificate subtraction, preserve no-crossing and null closure, and still produce a non-fresh admissible same-scope counter-boundary survivor outside the fixed-domain route matrix. Merely invoking hair, nonstationarity, dynamical evolution, a richer metric representative, or a different exterior presentation does not meet that burden unless it supplies standing-bearing re-fixation or explicitly changes the target.

Theorem 17.1 (Counterexample normal form). A successful challenge to the Counter-Boundary Universe Theorem must do at least one of the following:

1. deny the certified parent black-hole target;
2. break the kernel, AMetric boundary, null-regime, exhaustion, no-crossing, or homology import;
3. show that the antecedents are jointly inconsistent;
4. define a governance-relevant equivalence relation, compatible with the same boundary-pair target, under which FreshGov_{CB} is not unique;
5. show that the boundary-status burden is invalid, so that B_A^* can be a positive endpoint-status classification while imposing no counter-boundary status burden despite fixed-domain bivalence and endpoint classification;
6. exhibit a boundary-only terminality route that preserves positive endpoint-status, denies counter-boundary classification, and avoids collapse into inert silence, admitted nullity, mute-null-as-classification, fresh internal governed-domain status, or richer-scope governance;
7. show that the fixed-domain locus matrix omits a same-scope admissibility-relevant locus;
8. exhibit a same-scope admissible counter-boundary status outside the fixed-domain locus matrix and route register;
9. exhibit a crossing relation that preserves parent-scope standing without violating AMetric non-transmissivity;
10. show that role-level knowability has been overpromoted into inherited populated content;
11. show that downstream non-inheritance has been overpromoted into unconstrained downstream arbitrariness, or conversely that parameter-level variation alters a standing-bearing role ledger while the theorem treats it as skin;
12. for the life-carrier subsection specifically, deny the fixed-domain life antecedents, break standing localization, supply a standing-positive carrier without boundary, identity-memory, or local repair, supply a viability functional carrying standing-bearing force not reducible to $\Sigma = R_e - D$, supply non-local standing outside the failure-family exhaustion, or show a proper resolved subcarrier of a life carrier remains standing-positive.

Absent one of these, the objection is a scope change, skin, inherited-continuation request, selector request, mechanism preference, request for a physical production story, or demand for populated contents outside the theorem target.

18 Main Theorem Synthesis

The theorem ladder is:

1. Non-degenerate construction forces the kernel ($\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}$).
2. The admissibility interface is AMetric, bivalent, unique, and standing-conserving.

3. Same-scope repair, totalization, selector use, boundary metricity, boundary probability, primitive time, and multiple interiors are closed.
4. Total vacuity cannot be an admissibility-bearing regime.
5. Horizon isolation fixes a parent reference-standing burden; genuine exterior witness relations subtract only what they re-fix.
6. The unresolved nonempty residue has residual coherence without old-scope standing.
7. At maximum governance-depth, false endpoint classes exhaust.
8. The endpoint is B_A^* , a New AMetric Boundary.
9. $\text{PANF}(B_A^*) = \text{PANF}(B_A^0) = \text{aANF}$, with token distinction and no inheritance.
10. Nothing old-scope crosses.
11. Boundary-only terminality, nullity, and degenerate alternatives are closed.
12. Therefore a fresh internal governed domain—a constraint-forced universe in the AASC sense—is the unique non-degenerate counter-boundary status-class up to $\equiv_{\text{gov},CB}$.
13. If further target classes are fixed, constraint symmetry forces role-exact recurrence at the realized-interior, metric-projection, and realized-standard-physics structural levels; downstream contents remain non-inherited but internally certified.

The theorem also has a constraint-symmetry consequence with strict target discipline. If the fresh internal governed-domain status-class is represented under realized-physical target fixation, the forced RPI role package recurs. If minimal Lorentzian metric projection or realized-standard-physics target classes are fixed, their corresponding structural forms recur conditionally. If fixed-domain life antecedents are additionally fixed, minimal life-carrier structure/status recurs as a downstream target-class corollary. These are conditional structural recurrences, not inherited contents and not unconditional additions to the black-hole theorem. Downstream parameters, histories, records, occupants, life-carrier status, intelligence, and technology are non-inherited and require internal certification or further target-class antecedents.

Theorem 18.1 (Final synthesis). A certified AASC black-hole unresolved residue at maximum governance-depth does not terminate in singularity, storage, transfer, bounce, wormhole, inherited continuation, boundary-only terminality, admitted nullity, probability, quantum uncertainty, primitive time, metric passage, or ungoverned residue. It exposes a New AMetric Boundary. Across that boundary nothing old-scope crosses. Because boundary-only terminality, admitted nullity, and degenerate alternatives are closed, the boundary-pair register contains a unique non-degenerate counter-boundary status-class up to $\equiv_{\text{gov},CB}$, represented by a token-distinct fresh internal governed domain satisfying the AASC kernel and AMetric normal form. This is the precise AASC form of the controlled informal claim: a new constraint regime is forced, not inherited or causally produced. If further target classes are fixed, their role architectures recur by constraint symmetry; their concrete contents are not inherited from the parent and must be internally certified.

The theorem has a further constraint-symmetry consequence. If the fresh internal governed domain is instantiated as a realized physical interior, its role architecture is not freely variable: the same minimal physical-interior roles recur up to admissible equivalence. If further target classes are fixed, such as the minimal Lorentzian metric projection or the realized-standard-physics structural envelope, their corresponding structural forms recur conditionally. This does not transmit parent laws, records, constants, histories, or empirical contents. It says that any such downstream contents must be freshly certified inside the new domain.

Arc handoff summary

The capstone closes the arc at governance-status level. It does not add a mechanism, select a token cosmology, transmit parent law, transfer energy, preserve parent information, or populate a successor. Its final result is the unique non-degenerate counter-boundary status-class under the fixed antecedents: a fresh internal governed domain by constraint recurrence. This is the terminal claim of the series, not a premise of any earlier paper.

19 Conclusion

Series-position conclusion. This capstone is the final step of the arc rather than a premise for the earlier manuscripts. Its result depends on the completed chain from licensed black-hole role typing through exterior obstruction, standingless residual coherence, and AMetric homology. The controlled export of the whole sequence is fresh internal governed-domain status under the certified endpoint antecedents, with no inheritance or populated cosmography.

The theorem result is exact at its AASC status level: a certified AASC black-hole endpoint forces fresh internal governed-domain status in the AASC sense. It does not transmit old-scope standing, law, information, energy/source ledgers, metricity, clock structure, causal chains, fields, particle identity, memory, records, parameters, or operator domains. The no-crossing result fixes the claim's exact form as constraint recurrence rather than transmission.

The theorem also does not produce an inherited universe, a copied parameter table, a biological lineage, or a populated successor. It proves fresh internal governed-domain status. Constraint symmetry then explains why such a domain, once a target class is fixed, is not free to be radically arbitrary: the same minimal role obligations recur. Realized physical interiors, metric projections, standard-physics structural envelopes, and further downstream specializations recur only under their own audited antecedents. Concrete constants, histories, records, occupants, intelligence, and technology remain non-inherited and internally certified.

A New AMetric Boundary exposed at maximum governance-depth has the same complete AMetric normal-form profile as the origin-side licensing boundary:

$$B_A^\star \neq B_A^0, \quad \text{PANF}(B_A^\star) = \text{PANF}(B_A^0) = \mathbf{aANF}.$$

The Homology paper proved that profile equality and stopped there. The present theorem adds the next closure step. Once boundary-only terminality, admitted nullity, ungoverned residue, inherited continuation, probabilistic selection, primitive time, metric passage, causal production, selector authority, same-scope repair, and multiple rival interiors are closed, the boundary-pair classification register has a unique non-degenerate status-class: fresh internal governed domainhood up to $\equiv_{\text{gov}, CB}$.

Nothing crosses. That is the theorem's positive condition. Since inheritance is inadmissible, the recurrence cannot be transmissive, causal, or genealogical. A certified AASC black-hole endpoint cannot transmit a universe, inherit a universe, copy a universe, continue a universe, or cause a universe by parent-scope dynamics. It exposes the same AMetric licensing condition under which any non-degenerate governed interior is possible. Since the AASC kernel gives the minimal regime-invariant conditions for admissibility, standing, reference, and irreversibility, the homologous endpoint condition forces a fresh internal governed domain.

The result also blocks the opposite misreading: non-inheritance is not downstream arbitrariness. A fresh domain cannot obtain standing-bearing parameters, histories, records, law tokens, or empirical

contents by parent transfer, but neither can it obtain them by unconstrained stipulation. If such contents stand, they stand only through the new domain's own admissibility, reference, quotient, witness, calibration, and no-repair discipline.

In this precise sense, black holes make universes: certified AASC endpoints expose the same minimal licensing condition that forces non-degenerate governed interiority. Nothing is inherited; that is the condition under which the result has its exact form. The recurrence is role-exact at the forced-architecture level and non-inherited at the concrete-content level. Fresh universes are role-equivalent by constraint at the forced-architecture level, not copied by transmission. What is unique is the governance class and its target-fixed structural corridors; concrete contents, numerical parameters, and histories remain internal to the fresh domain or to later target-class certificates.

20 Formal Dictionary

Symbol or term	Meaning
D_0	Parent governed regime under classification.
D_1	Representative notation for a token-distinct member of the fresh internal governed-domain class; not a selected token cosmology.
B_H	Horizon-generated parent reference-standing burden.
$W_{\text{ext}}(B_H)$	Portion of B_H resolved by genuine exterior witness relations.
R_H	Unresolved horizon-isolated residue, $B_H \setminus W_{\text{ext}}(B_H)$.
B_A^0	Origin-side AMetric boundary token.
B_A^*	Endpoint-side New AMetric Boundary token.
$\text{PANF}(B)$	Complete profile of the AMetric boundary token B .
aANF	AMetric normal form.
No crossing	No parent-scope standing-bearing item has a same-item image across B_A^* .
Fresh internal governed-domain status	Kernel instantiation under AMetric normal form without inheritance.
Internal governed domain	A non-null domain-status instantiating its own admissibility gate, standing relation, reference-fixation discipline, irreversibility/no-repair structure, and AMetric licensing token.
AASC universe	An internal governed domain at the universe-governance register; realized-physical roles are forced when the realized-physical target class is fixed.
Constraint-forced universe	A fresh internal governed domain forced by kernel minimality and AMetric boundary discipline after non-fresh alternatives close.
Structural identity without inheritance	Role-architectural sameness caused by minimal constraints, not by crossed contents, copied law, inherited metricity, or genealogical continuity.
$\equiv_{\text{gov}, CB}$	Counter-boundary governance equivalence: sameness of admissibility-bearing governance coordinates, not sameness of token universe or populated content.
Regime occupation	Non-null governed domain-status.

Status-existence	Typed admissibility classification in the boundary-pair register.
Constraint identity	Independent instantiation of the same minimal role architecture required by the same non-degeneracy conditions, without inheritance or token identity.
Role-exact recurrence	Recurrence of a forced role package up to admissible equivalence under a matched target class.
Internal certification	Domain-local licensing of downstream contents by admissibility, standing, reference, quotient, witness, calibration, and no-repair discipline.
Populated-existence	Independently licensed concrete internal occupants, histories, fields, law tokens, parameters, records, or observers.
Residual coherence	Role-indexed organization after standing transport failure, without full standing.
Maximum governance-depth	Endpoint condition where residual old-scope role support for metric/source/temporal/causal/operator articulation exhausts.

21 Claim-Status Ledger

Claim	Status	Discipline
B_A^* is exposed at maximum governance-depth for nonempty R_H .	Imported	New AMetric Boundary endpoint theorem.
Nothing old-scope crosses B_A^* .	Imported/used	Absolute no-crossing theorem.
Boundary-pair comparison scope is the theorem object.	Proved here	Classification scope, not transport domain.
Admitted nullity is unavailable.	Imported	Null-regime closure.
AMetric homology holds.	Imported	$\text{PANF}(B_A^*) = \text{PANF}(B_A^0) = \text{aANF}$.
Completed closure register assumes the conclusion.	Denied	The Completed Closure Register Lemma records route-register closure; it does not select the fresh internal governed-domain survivor as a premise.
Fresh internal governed domain-hood is the unique counter-boundary status-class.	Proved as status-class up to $\equiv_{\text{gov}, CB}$	Boundary-status burden plus route-register exhaustiveness plus null closure plus no-crossing plus homologous boundary licensing plus kernel minimality.
Certified black-hole endpoints force fresh internal governed-domain status in the AASC sense.	Proved under antecedents	“Make” means force fresh internal governed domain-status by constraint recurrence, not causal production or inheritance.

The theorem is universal over certified post-subtraction targets.	Locked		Astronomical identification alone is not the endpoint handoff; a genuine counterexample must preserve the unresolved residue after quotient/certificate subtraction while producing a same-scope survivor outside route closure.
$\equiv_{\text{gov},CB}$ hides real differences.	Denied		It preserves all standing-relevant governance coordinates and quotients only downstream token/content differences.
The new universe is role-equivalent because constraints are minimal.	Proved	at	Identity is role-architectural, not token identity, content identity, or parameter identity.
A unique token universe is proved.	Denied		The theorem proves status-existence of a governance-equivalence class, not token selection.
No inheritance is a limitation on the universe claim.	Denied		No inheritance is the positive condition that makes the result fresh rather than transmissive.
Populated-existence is proved.	Denied		Concrete contents require independent internal licensing.
D_1 names a selected concrete cosmology.	Denied		D_1 is representative notation for the fresh internal governed-domain class.
Parent contents populate D_1 .	Denied		Would violate no-crossing.
Concrete laws of D_1 are parent-derived.	Denied		Would require inherited governance.
Role architecture of D_1 is knowable.	Proved	conditionally	Kernel always; realized-interior and SM layers only under target fixation; life-carrier layer only under fixed-domain life antecedents.
Numerical constants recur by inheritance.	Denied		Parameters require internal anchors and certificates.
The theorem proves cyclic cosmology.	Denied		Homology is structural, not temporal.
The theorem proves a baby-universe mechanism.	Denied		No causal production or inherited population; the positive theorem is fresh constraint-forced universe-status.
Fresh interiors are role-equivalent by inheritance.	Denied		Role identity is constraint identity, not crossing or genealogy.

Fresh interiors are role-exact under target fixation.	Proved conditionally	Kernel always; realized-interior, Einstein-projection, and Standard-Model-structural layers only under their target-class antecedents; minimal life-carrier recurrence requires additional fixed-domain life antecedents.
Parameter values are unconstrained because they are non-inherited.	Denied	Non-inheritance blocks parent supply; internal certification remains required.
Raw numerical displays are the real content of the fresh regime.	Denied	Raw displays are skin unless promoted by internal witness, quotient, calibration, or exactness discipline.
Standard-Model-like structure recurs in every fresh domain automatically.	Denied	It recurs when the realized-standard-physics target class is fixed.
Radical downstream variation is freely available.	Denied	Variation is available only where it preserves the relevant standing-bearing role ledgers.

22 Proof Dependency Graph in Text Form

Kernel necessity

- > Structure of admissibility
 - > AMetric boundary normal form
 - > no selector / no parameter / no metric / no probability
 - > bivalence / standing identity / standing conservation
 - > unique admissible interior
- > Fixed-domain exhaustion and operator closure
- > Null-regime closure
- > Realized physical interior (conditional target layer)
- > Einstein projection (licensed parent metric projection)
 - > Black-hole horizon isolation
 - > exterior witness subtraction
 - > unresolved residue RH
 - > standingless interior / residual coherence
 - > maximum governance-depth
 - > New AMetric Boundary B_A^*
 - > absolute no-crossing
 - > AMetric normal-form homology
 - > Constraint-Forced Universe Theorem
 - > constraint symmetry under target fixation
 - > internal certification

23 Endpoint Route-Exhaustion Matrix

| Route | Required role | Failure |
|----------------------------------|--|---|
| Singularity | Metric diagnostic authority. | Metric role support exhausted at maximum governance-depth. |
| Remnant | Standing-bearing storage carrier. | Carrier identity and continuation unavailable for R_H . |
| Archive | Record identity and retrieval path. | Record standing requires witness and quotient identity. |
| Information sink | Primitive information cargo. | Information requires carrier, encoding, reference, witness, standing. |
| Bounce | Time, causal transition, metric continuity. | These are unavailable at AMetric endpoint. |
| Wormhole | Passage, throat metric, causal relation. | Requires old-scope metric/causal bridge. |
| Transfer | Source ledger and cross-boundary target. | Requires shared ledger and transport rule. |
| Null endpoint | Admissible vacuity. | Closed by null-regime theorem. |
| Probability selector | Measure over alternatives. | Probability is interior, not boundary authority. |
| Quantum uncertainty | Quantum state/amplitude package. | Downstream of governance. |
| Same-scope repair | Operator totalization. | Closed by no-repair and operator exhaustion. |
| Inherited law | Cross-boundary authority. | Violates no-crossing. |
| Fresh constraint-forced universe | Kernel under AMetric normal form; fresh internal governed domain-status. | Survives. |

24 Forced-Without-Crossing Ledger

| Layer | Status | Forced | Not inherited |
|----------------------------|---|--|--|
| Kernel | Unconditional | Admissibility, standing, reference, irreversibility, bivalence, no repair, standing conservation. | Parent acts, standing tokens, records. |
| AMetric | Unconditional | No primitive metric, time, probability, selector, representative, inherited law, operator. | Parent metric, clocks, causal order, law package. |
| Realized physical interior | Conditional on realized-physical target | Locality, metric licensing, fields, particles, gauge, mass/inertia, vacuum, time, causality, records as roles. | Concrete fields, metric token, occupants, histories. |

| | | | |
|----------------------------------|--|--|---|
| Standard Model structural | Conditional on realized-standard-physics target | Gauge product, representations, Higgs/bridge, Yukawa slots, charge lattice, anomaly neutrality, vacuum compatibility. | Coupling values, masses, mixing values, empirical parameter table. |
| Minimal life-carrier target | Further conditional on fixed-domain life antecedents | Minimal bounded carrier form under local repair, retained identity-memory, standing margin, localization, and finite standing-fixation interval. | Inherited life, molecular substrate, organisms, evolution, intelligence, technology, abundance. |
| Constraint symmetry | Conditional on matched target class | Role-exact recurrence up to admissible equivalence. | Token identity, inherited contents, copied parameter table. |
| Parameter record / history level | Non-inherited but internally constrained | Standing-bearing parameter slots, empirical records, calibration ledgers, law tokens, and histories require internal certification. | Parent values, parent records, parent histories, parent law tokens. |

25 Black-Hole Residue Certification Protocol

For a declared black-hole target, apply the following certification protocol.

1. Fix the parent exterior reconstruction scope: exterior standing quotient, witness algebra, continuation relation, parameter ledger, thermodynamic ledger, and certificate status.
2. Identify horizon-generated reference-standing burdens B_H .
3. Identify genuine exterior witness relations and compute $W_{\text{ext}}(B_H)$.
4. Form $R_H = B_H \setminus W_{\text{ext}}(B_H)$.
5. If $R_H = \emptyset$, endpoint residue theorem does not activate.
6. If $R_H \neq \emptyset$, classify residual coherence and governance deficiency.
7. Check whether independent re-governance is declared before metric support exhaustion.
8. If no such governance is declared and residual support exhausts, classify endpoint routes.
9. If singularity, remnant, storage, transfer, bounce, wormhole, boundary-only terminality, probability, time, metric bridge, or nullity is proposed, apply the false-route matrix.
10. If all false routes close, report B_A^* and no-crossing.
11. Apply the Counter-Boundary Universe Theorem only after B_A^* , no-crossing, null closure, and route exhaustion are fixed.

26 Reconciliation with the Conservative Endpoint Posture

The conservative sentence from the endpoint paper was:

AASC endpoint analysis of D_0 cannot establish the contents, laws, population, metricity, or concrete instantiation of any post-boundary domain.

The sentence remains true if read as a no-inheritance and no-populated-content claim. It is not the strongest possible closure statement once nullity and degenerate routes are closed. The present theorem

adds:

AASC endpoint analysis plus null-route closure, AMetric homology, route exhaustion, and kernel minimality forces status-existence of a fresh internal governed domain: a constraint-forced universe in the AASC sense, while still denying inherited contents, laws, population, metricity, parameters, and parent-derived empirical facts.

There is no contradiction. The conservative endpoint sentence blocks a crossing or populated-successor over-extension. The new theorem blocks the opposite underclaim that no-crossing leaves only arbitrary silence or nullity. No-inheritance is not the absence of a result; it is the exact condition under which the result becomes fresh rather than transmissive.

27 Extended Objection Ledger

| Objection | Reply |
|--|--|
| A new regime with no inherited laws is arbitrary. | Its concrete contents are not parent-derived, but its role architecture is not arbitrary. Kernel and target-class necessities force it once non-fresh alternatives close. |
| Formal endpoint slogan. | Correct only as shorthand for: certified endpoints force fresh internal governed domains by constraint recurrence. Incorrect if read as inheritance, causal production, or baby-universe tunneling. |
| Without a causal relation, why call it counter-boundary? | Counter-boundary is a classification relative to the boundary token, not a causal relation. |
| If no parent content crosses, endpoint-status derivability is itself crossing. | Endpoint-status derivability is a parent-side classification of R_H 's closure, not a value imported from D_1 . |
| Homology carries no force. | Homology plus null-route closure, no-crossing, and kernel minimality has constraint force: it blocks arbitrary nullity and inherited continuation while forcing fresh internal governed domain-status. |
| Homology secretly asserts recurrence of our universe. | No. It asserts equality of AMetric role profile. The present theorem adds recurrence of the universe-licensing condition, not recurrence of parent contents. |
| A probability distribution over regimes could exist externally. | That is a third-domain structure with its own governance; it is not boundary authority in the parent scope. |
| The theorem turns absence into existence. | It closes admitted nullity and ungoverned residue inside the theorem target. Fresh internal governed domainhood is the unique non-degenerate status-class up to $\equiv_{\text{gov},CB}$, not a conjuring of populated contents from absence. |
| The theorem is metaphysical. | It is an AASC fixed-domain route-exhaustion theorem. Its proof burden is target, imports, routes, and status-class classification. |

| | |
|---|---|
| The result depends on words like governance. | The vocabulary can be replaced only if the roles are preserved: admissibility, standing, reference, irreversibility, no repair, no selector, no crossing. |
| This collapses no-inheritance into sameness. | No. No-inheritance blocks parent transmission. Constraint symmetry says independently instantiated non-degenerate domains solve the same minimal role problem. The sameness is necessity-level, not genealogical. |
| If parameters can differ, the interiors are not structurally identical. | Parameter variation does not defeat role equivalence unless it changes a standing-bearing role ledger. Raw numerical displays are skin unless internally certified; structural identity is evaluated at the kernel, realized-interior, metric-projection, and structural-envelope levels. |
| The paper claims every fresh universe has the Standard Model. | No. The claim is conditional. If the fresh domain is fixed as a realized-standard-physics target, the Standard Model structural corridor recurs. The black-hole theorem alone proves fresh internal governed-domain status. |
| Parameters are arbitrary because nothing crosses. | No. Nothing crosses from the parent. Any parameter or record inside the fresh domain must be internally licensed by that domain's own admissibility and standing conditions. |
| The theorem overclaims life. | The life-carrier claim is further conditional on fixed-domain life antecedents. It is not biological inheritance, abiogenesis probability, molecular substrate, intelligence, technology, detectability, or abundance. |
| The theorem infers life from black-hole endpoints. | No. The black-hole theorem proves fresh internal governed-domain status. Life-carrier recurrence is a further conditional target-class result. |
| The life layer is speculative biology. | The life-carrier import is a fixed-domain standing-localization theorem using local repair, retained identity-memory, standing margin, localization, and strict minimal carrier descent; molecular substrate and empirical abundance are outside the claim. |

28 Source-Use Ledger

| Source manuscript | Use in present paper |
|--|--|
| Non-Degenerate Construction and the Kernel of Admissibility | Kernel roles, no lower generator, AMetric boundary consequences, no same-act repair. |
| The Structure of Admissibility | AMetricity, bivalence, standing identity, uniqueness, conservation. |
| The AMetric Boundary as a Unique Regime-Invariant Constraint | No selector, no parameter, local route exhaustion, export discipline. |
| The Structural Inadmissibility of the Null Regime | Closure of admitted nullity and mute-nullity discipline. |
| Unus Solus Possibilis Est | Uniqueness of the admissible interior and no multiple same-scope interiors. |

| | |
|--|--|
| Closure by Exhaustion for Same-Scope Operators | No repair, totalization, selector, completion, or same-scope operator escape. |
| A Fixed-Domain Exhaustion Theorem | Theoremic, non-enumerative exhaustion method. |
| The Arrow of Time as Standing Monotonicity | No primitive temporal direction and no temporal backflow. |
| The Physical Clock as a Quotient | Clock-use non-primitivity and no primitive external clock selection. |
| Anchor, Tensor, and Skin | Role decomposition; no fourth same-domain role; skin cannot carry boundary authority. |
| Metricity as Licensed Description | No primitive metricity and no metric backflow. |
| The Realized Physical Interior | Conditional role package for realized-interior recurrence; does not prove populated successor or inherited contents. |
| The Standalone Standard Model Structure from First Principles | Conditional standard-physics structural corridor and parameter-status discipline; does not close the full empirical parameter table. |
| Standing Localization and Minimal Life Carriers | Supports only conditional life-carrier recurrence under fixed-domain life antecedents: local repair, retained identity-memory, standing margin, localization, strict minimality, and finite standing-fixation interval; does not prove inherited life, substrate, intelligence, technology, SETI/Fermi claims, or abundance. |
| Einstein Dynamics as the Unique Minimal First-Response Metric Projection | Conditional metric-form recurrence under faithful minimal Lorentzian target; does not derive numerical κ , Λ , or all gravitational dynamics. |
| Black Holes as Licensed Limit Regimes | Horizon isolation, exterior witness algebra, no-totalization. |
| Standingless Interiors and Residual Coherence | Residual coherence, governance deficiency, no horizon-as-AMetric-reset. |
| The New AMetric Boundary Endpoint of Black Holes | Endpoint closure, no inheritance, absolute no-crossing. |
| AMetric Normal-Form Homology | Origin-endpoint AMetric profile equality and no cyclic/transmissive reading. |
| Early Metric-Interior Transport Closure | Origin-side metric-interior transport discipline and no primitive inflationary backflow. |

29 Short Formal Proof

Given. $R_H \neq \emptyset$, maximum governance-depth, B_A^* , boundary-status burden, no-crossing, null closure, and $\text{PANF}(B_A^*) = \text{aANF}$.

Suppose. A non-fresh alternative X is the counter-boundary status.

Cases.

1. If X is inherited, it violates no-crossing.

2. If X is same-regime continuation, it requires old-scope standing transport, which is undefined.
3. If X is ungoverned, it lacks the kernel and is not a non-degenerate regime.
4. If X is admitted nullity, it is closed by the null-regime theorem.
5. If X is mute nullity, it has no classification force; if used to classify, it is not mute.
6. If X is boundary-only terminality, it is either inert, null-closed, or reclassified as a fresh internal governed domain or richer scope.
7. If X is selected by probability, time, quantum amplitude, metric bridge, causal passage, or representative privilege, it imports downstream or forbidden boundary authority.
8. If X is a repaired or totalized old-scope operator, it violates operator closure and no repair.
9. If X is one of multiple same-scope admissible interiors, it violates uniqueness.
10. If X is a richer third-domain construction, it is not same-scope crossing and must instantiate its own governance.

All non-fresh alternatives close. Therefore the unique non-degenerate counter-boundary status-class up to $\equiv_{\text{gov},CB}$ is a fresh internal governed domain. By kernel necessity, it instantiates (**Adm**, **Stand**, **Ref**, **Irr**). By AMetric homology, its boundary normal form is **aANF**. By minimality, it is role-equivalent to any origin-licensed governed-interior architecture at the role level. By no-crossing, it is not inherited. QED.

30 Boundary of the Result

This theorem does not establish:

- that an astronomical label alone supplies the endpoint handoff;
- that every black hole leaves a nonempty unresolved residue;
- that any particular quantum-gravity proposal fails;
- that a metric, law package, particle inventory, or population exists inside D_1 ;
- that numerical constants or empirical parameters are inherited; admissible recurrence, if any, requires internal certification inside the fresh domain;
- that D_1 is causally produced;
- that cyclic cosmology is true;
- that parent information, energy, or law is conserved by transfer.

It does establish the fixed target-class result: if the antecedents hold, no old-scope crossing occurs, nullity and degenerate alternatives close, and fresh internal governed domainhood has status-existence as the unique non-degenerate status-class up to $\equiv_{\text{gov},CB}$. In controlled shorthand, the endpoint forces a fresh governed domain by constraint recurrence.

31 Richer-Scope Declaration Protocol

A richer-scope theory may be valuable, and the present theorem does not prohibit it. The protocol is simply that richer scope must be declared rather than smuggled as old-scope crossing.

A richer-scope declaration must specify:

1. the new domain carrier;
2. the admissibility gate of that domain;

3. its standing relation;
4. its reference-fixation method;
5. its irreversibility/no-repair discipline;
6. its relation, if any, to D_0 and D_1 , without treating that relation as parent-scope crossing;
7. whether it resolves R_H before endpoint closure or only classifies a third-domain comparison afterward.

A theory that supplies this package may deny an antecedent of the Counter-Boundary Universe Theorem. A theory that does not supply it is not a counterexample.

32 Language Classifier

| Phrase | Status | Replacement |
|---|---|---|
| Ordinary-language slogan. | Admissible only with the controlled AASC meaning. | Certified black-hole endpoints force fresh internal governed domains by constraint recurrence, not by inheritance or causal production. |
| The black hole creates a universe from parent contents. | Rejected. | No parent-scope contents cross; the new domain is fresh. |
| A universe comes after the boundary. | Temporal backflow. | A counter-boundary regime is classified relative to B_A^* , not temporally sequenced from D_0 . |
| Information goes into the new regime. | Crossing violation. | Parent information does not cross; any new information roles are internal to the fresh internal governed domain. |
| Energy is transferred. | Ledger backflow. | Parent energy/source ledgers do not cross; D_1 has its own source roles if realized. |
| The far side has the same laws. | Inheritance over-extension. | Role architecture recurs by constraint under target fixation; law tokens are not inherited. |
| The new universe is identical to the old one. | Ambiguous. | It is role-exact at the forced-architecture level under target fixation, not token-identical, content-identical, or historically continuous. |
| Nothing crosses, so nothing follows. | Rejected. | Nothing crosses; therefore inheritance fails, and fresh constraint-governed recurrence is forced once nullity and non-fresh alternatives close. |
| Parameters are free because nothing crosses. | Rejected. | Downstream contents are non-inherited but internally constrained. |

The Standard Model is Rejected.
copied into the new universe.

The boundary selects an Selector violation.
outcome.

The new regime is unknow- Too strong.
able.

The Standard Model structural corridor recurs only under realized-standard-physics target fixation and only up to admissible redescription.

No selector has AMetric boundary authority.

Concrete contents are not inherited; role architecture is knowable by necessity.

33 Referee Checklist

A referee checking the theorem should ask these questions in order.

1. Does the manuscript state that no-inheritance is the positive engine of the theorem?
2. Does it distinguish AMetric licensing condition from physical initial condition?
3. Does it define AASC universe-status as internal governed domainhood rather than inherited population?
4. Does it distinguish B_A^* , B_A^0 , and $B_A^{D_1}$?
5. Does $\equiv_{\text{gov}, CB}$ preserve all governance-relevant coordinates?
6. Does the route register close the fresh-but-non-universe route?
7. Does the manuscript state constraint symmetry without converting no-inheritance into genealogy?
8. Does it distinguish role-exact recurrence from parameter/content identity?
9. Does it state that downstream parameters, records, histories, and law tokens are non-inherited but internally certified?
10. Does it keep Standard-Model structural recurrence conditional on realized-standard-physics target fixation?
11. Has the parent target D_0 been certified?
12. Is R_H nonempty after genuine exterior witness subtraction?
13. Has maximum governance-depth been reached by the stated antecedent rather than assumed globally for all black holes?
14. Is B_A^* imported from the endpoint theorem rather than introduced locally?
15. Is no-crossing treated as load-bearing rather than rhetorical?
16. Is admitted nullity closed directly?
17. Are probability, quantum, temporal, metric, causal, selector, and repair routes each typed as downstream, prohibited, or richer-scope?
18. Is fresh internal governed domainhood distinguished from inherited governance and from causal production?
19. Is role-level knowability distinguished from populated contents?
20. Is the Standard Model layer stated only under target-class fixation?
21. Is the life-carrier layer stated only as a further conditional downstream specialization under fixed-domain life antecedents?
22. Does it prove the constraint-symmetry implication rather than leaving fresh interiors arbitrary?
23. Does it distinguish role-exact recurrence from inherited sameness?

24. Does it state that downstream parameters, records, histories, and law tokens are non-inherited but internally constrained?
25. Does it avoid saying that raw numerical displays are the real content unless internally certified?
26. Does any baby-universe language remain subordinate to the exact AASC claim: fresh internal governed domainhood by constraint recurrence?

34 Deletion Matrix

| Deleted component | First failure |
|--|---|
| No-crossing | Inherited successorhood becomes available and the theorem collapses into a transmission theory. |
| Null-regime closure | The status-class survivor step fails because admitted vacuity remains. |
| AMetric homology | The endpoint has no boundary normal-form constraint strong enough to force fresh internal governed domainhood. |
| Kernel necessity | Non-degenerate occupation loses the forced roles of admissibility, standing, reference, and irreversibility. |
| Operator closure | Same-scope repair or totalization could mimic crossing. |
| No primitive time/clock | Counter-boundary relation could be misread as temporal succession. |
| Metricity licensing | Metric bridge or singularity could be overpromoted into endpoint authority. |
| Unus/uniqueness | Multiple same-scope interiors could be asserted without role distinction. |
| Role-level/populated-content distinction | The theorem either overstates concrete contents or understates the constraint-forced universe result. |
| Minimal Constraint Identity Lemma | Role identity appears inherited or definitional rather than forced by kernel minimality. |
| AASC universe definition | “Universe” appears rhetorical or merely stipulative rather than theorem-typed. |
| Boundary-token discipline | Endpoint exposure risks being misread as boundary transmission. |
| Black-Holes-Make-Universes corollary | The public theorem claim remains underreported. |
| Fresh-but-non-universe route closure | A referee can concede fresh internal governed-domain status while denying universe-status. |
| Constraint Symmetry section | Fresh interiors appear either arbitrary or secretly inherited instead of role-exact by target-fixed non-degeneracy constraints. |
| Downstream non-arbitrariness proposition | Parameter, record, and history variation can be misread as unconstrained multiverse freedom. |
| Realized-interior recurrence proposition | The theorem fails to connect fresh universe-status to the RPI role package. |
| Standard-physics corridor proposition | Standard Model structural recurrence appears either automatic overclaim or absent implication. |

| | |
|-----------------------------------|---|
| Internal certification definition | Downstream contents have no domain-local standing test after no-crossing. |
|-----------------------------------|---|

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